Olera CareNavigator

Increasing Affordability and Accessibility of Senior Care for Dementia Family Caregivers

Abstract

Senior care is unaffordable for millions, yet a large percentage of elderly Americans miss out on $30 billion yearly in unclaimed financial aid because they are unaware that they qualify for assistance with food, housing, prescriptions, and healthcare or they do not know how to apply. This presents a high-impact opportunity to help millions with transformative innovation. Recent advances in artificial intelligence (AI) called Large Language Models (LLMs) could allow for mass eligibility-screening of elderly Americans and optimal resource-matching with underutilized local, state, or federal aid programs and other support programs. Novel applications of AI in the eldercare ecosystem have the potential to help caregivers especially for individuals with demanding conditions like Alzheimer's and related dementia (AD/ADRD) facing complex care demands and frequent access and affordability issues. In this proposed study, we aim to develop, evaluate, and deploy an AI-powered care planner technology to enable Americans to reclaim billions in underutilized aid and better access vital support resources. This collaborative project, involving industrial, academic, and community partners, is oriented around three specific aims: (Specific Aim 1) **Development of novel specialized AI agents for elder care planning.** Three categories of AI “agents" will be engineered for specialized care planning functionalities including: needs assessment agents, resource & aid matching agents, and personalized care planning agents. (Specific Aim 2): **Integration of multi-agent network & preliminary usability testing of a novel intelligent user interface (UI) among AD/ADRD caregivers.** The agents will be integrated into a Multi-Agent Network and intelligent UI. The UI will be evaluated among a pilot group of AD/ADRD family caregivers (n=25) using the User Experience in Intelligent Environments (UXIE) framework to test for its acceptance and usability. Following a Build-Measure-Learn approach, we will iteratively refine the UI through survey data, platform usage data, and interview feedback. (Specific Aim 3): **Evaluation of the user experience and caregiver perceptions of novel AI-powered care planning technology among AD/ADRD caregivers with diverse backgrounds.** The network will be evaluated for usability, acceptance, and caregiving experience among AD/ADRD family caregivers (n=200) with diverse backgrounds regarding demographic characteristics and health needs factors. The modified Technology Acceptance Survey (TAS), the modified Mobile Application Rating Scale (MARS), Caregiver Self-Efficacy Scale (CSES-8), and the positive and negative appraisals of caregiving (PANAC) will be used as key quantitative survey tools to evaluate the technology acceptance and usability, and the caregiving experience in decision making before and after using the technology. At Phase IIB conclusion, the care-planner technology will be ready for service, commercially viable, and prepared for nationwide scaling.

PROJECT NARRATIVE

This project addresses one of the most significant issues with the U.S. elder care system: the prohibitively high senior care costs and barriers to accessing care. It does so through the development and evaluation of a novel AI-powered care planning technology designed to help caregivers of older adults in need of care, particularly those with AD/ADRD, find underutilized financial aid resources and personalized guidance to accessing and afford available support systems on the local, state, and federal levels. The project builds on a previous Fast-Track SBIR project developing a web-based caregiver support platform (The Olera Platform) that connects families to local financial, legal, and care resources by applying Large Language Models (LLMs), a novel class of AI, and generative AI techniques in an innovative approach to senior care planning.

SPECIFIC AIMS

Background. Senior care is unaffordable for millions,[1,2] yet a large percentage of elderly Americans miss out on $30 billion yearly in unclaimed financial aid because they don't know how to apply or are unaware they qualify for assistance with food, housing, prescriptions, and healthcare.[3] This presents a high-impact opportunity to help millions by leveraging emerging technology advancements. Recent breakthroughs in artificial intelligence (AI) called Large Language Models (LLMs) could allow for mass eligibility screening of elderly Americans and optimal resource-matching with underutilized local, state, or federal aid programs or lesser-known support like volunteer groups or community service organizations. Further, AI technology could deliver on-demand guidance to help with aid applications or even generate on-request personalized information about relevant support services like home care or medical transportation. Novel applications of AI in the eldercare ecosystem like this have the potential to drastically improve how we currently approach resource-matching and service coordination for the elderly, especially for individuals with demanding conditions like Alzheimer's and related dementia (AD/ADRD) facing complex care demands and frequent access and affordability issues.[4] The impact could be substantial in creating a more resource-connected aging population in the US and ultimately lead to improved well-being and health outcomes across the nation.

SBIR Objective: We aim to develop, evaluate, and deploy the Olera CareNavigator – an AI-powered care planner technology to enable Americans to reclaim billions in underutilized aid and better access vital support resources. This Phase IIB end-product will be service-ready, commercially viable, and positioned to scale nationwide to help families through comprehensive resource-matching and personalized guidance on accessing and affording care services.

Specific Aim 1: Development of Novel Specialized AI Agents for Elder Care Planning. Three categories of AI "agents" will be engineered for specialized care planning functionalities including (1) needs assessment agents, (2) resource & aid matching agents, and (3) personalized care planning agents. Outputs generated by agents will be optimized by licensed clinical social workers (LCSW) through a machine learning technique called reinforcement learning with human feedback (RLHF). To generate accurate responses to caregiver inputs, agents leverage a novel elder-focused LLM that takes advantage of two recent innovations in model training: Retrieval-Augmented Generation (RAG) and Parameter- Efficient Fine-Tuning (PEFT).[5]

Specific Aim 2: Integration of Multi-Agent Network & Preliminary Usability Testing of a Novel Intelligent User Interface (UI) Among AD/ADRD Caregivers. The agents will be integrated into a Multi-Agent Network and intelligent User Interface (UI) novel in its ability to simulate dynamic human conversation crafted using generative AI techniques.[6-8] The UI will be evaluated among a group of AD/ADRD family caregivers (n=25) using the User Experience in Intelligent Environments (UXIE) framework to test for its acceptance and usability. Following a Build-Measure-Learn approach,[9] we will iteratively refine the UI through survey data, platform usage data, and interview feedback.

Specific Aim 3: Evaluation of the User Experience and Caregiver Perceptions of Novel AI-Powered Care Planning Technology Among AD/ADRD Caregivers with Diverse Backgrounds. The network will be evaluated for usability among AD/ADRD family caregivers (n=200) with diverse backgrounds. The modified Technology Acceptance Survey (TAS), the modified Mobile Application Rating Scale (MARS), Caregiver Self-Efficacy Scale (CSES-8),[10] and the positive and negative appraisals of caregiving (PANAC)[11] will be used as key quantitative survey tools to evaluate the technology acceptance and usability, and the caregiving experience in decision making before and after using the technology.

Vision: At Phase IIB conclusion, the Olera CareNavigator will be ready for service, commercially viable, and prepared for nationwide scaling. With our diverse interdisciplinary team, we aim to help Americans access billions in underutilized aid, better connect with vital support resources, and ultimately positively impact the well-being of 61 million with disabilities and their 53 million family caregivers[12] (11 million caring for AD/ADRD).[13]

Research Plan

SIGNIFICANCE

Challenges with the US elder care ecosystem: effect on family caregivers. Family caregivers, particularly those assisting older adults with AD/ADRD often face financial strain, job disruptions, and the emotional toll due to high expenses, lack of guidance, and demanding caregiving duties.[1-4] After closely working with over 200 families navigating AD/ADRD in our previous Fast Track SBIR project, we have witnessed an urgent need for better support systems for families looking for care. The subsequent summary outlines the two most prominent challenges identified in a formal thematic analysis of challenges AD/ADRD caregivers face conducted during our preliminary research:[5]

Challenge #1: Elder care is unaffordable for most Americans. The costs of elder care, particularly for individuals living with AD/ADRD, have risen significantly over the past few decades to the point where the average American cannot afford care.[4] In the U.S. the average annual cost of memory care is $83,220 while full-time home care costs $62,400 on average.[6] On the other hand, the average annual retirement income for a retired worker in the United States is roughly $20,000 according to the U.S. Social Security Administration, leading many to rely on financial assistance to pay for elder care.[7] Rising labor rates, a labor shortage, and increased prices of medical goods are among the reasons for an explosion in the cost of care. These challenges are likely to intensify in the coming decades as the aging population grows.[8]

There is a significant gap between the needs of the majority of Americans and the available support systems,[9] necessitating urgent attention. Medicare, the public healthcare program that the majority of older Americans use,[10] does not cover the cost of long-term care,[11] and Medicaid only applies to a segment of the population with minimal assets.[12] According to the Centers for Medicare & Medicaid Services (CMS), less than 20% of the U.S. population qualifies for Medicaid, and waivers for extended long-term care home and community-based services are insufficient.[13]

Challenge #2: The elder care ecosystem is fragmented, bureaucratic, and difficult to navigate. Millions of elderly Americans miss out on $30 billion yearly in unclaimed financial aid because they don't know how to apply or are unaware they qualify for assistance with food, housing, prescriptions, and healthcare (Figure 1).[14] Determining appropriate care services and aid is challenging for many.[5,15] It requires assessing needs and then matching to local, state, and federal resources which vary widely and data is not centralized.[16] It then requires applications, phone calls, and time to ultimately initiate a service or utilize an aid program.[17] Unfortunately, many are lost to follow up or never initiate support systems, and needs go unmet.[18] For example, nearly half of seniors who qualify for SNAP do not enroll because they are unaware of the program, if they qualify, and how to apply. That's an estimated 5 million people who are missing out on food assistance, amounting to $6.3 billion each year.[19] Initiating the right care services requires navigating a complex ecosystem with many components like government programs, insurance, and out-of-pocket costs.[20]

Figure 1: Unclaimed financial aid by older adults and family caregivers

While care coordinators like social workers or case managers can provide valuable personalized guidance,[17] their services are often restricted by workforce shortages[21,22] and research on expanding direct social work in geriatrics is scarce, but currently a priority according to experts in the field as a possible aid in combating caregiver burden.[23] Eldercare finder platforms online are available for information on senior living, such as A Place For Mom, Care.com, and SeniorAdvisor.com. However, these services are generalized and not designed to offer personalized guidance, but rather largely aim to match families with private-pay elder care options, which are unaffordable to a large percentage of the population. Unfortunately, there are not many accessible, innovative solutions focused on finding and securing affordable care suitable for most families.

Introducing the Olera CareNavigator: An AI-Powered Care Planner to Improve Accessibility to Affordable Eldercare. We aim to apply Large Language Models (LLMs), a novel class of AI, to improve elder care accessibility & affordability by helping individuals 1) find affordable care solutions and 2) better utilize resources and aid programs. Our approach involves using LLMs to create three care planning “AI agents” (Figure 2) specialized in assessing functional needs, identifying relevant support resources, and generating personalized guidance in accessing and utilizing support systems.

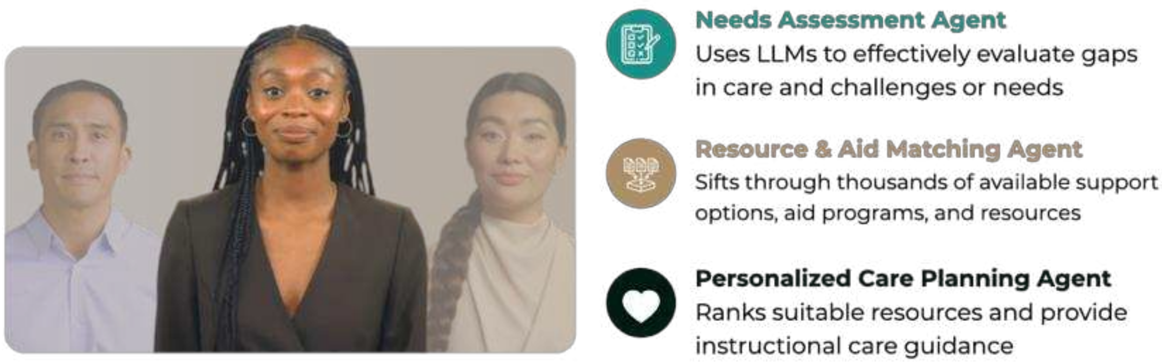

Figure 2: An illustration of the three-agent model that constitute the Olera CareNavigator

The proposed project builds on a strong technology foundation and robust community of AD/ADRD caregivers that was established over the past three years by Olera, Inc. during a related Fast-Track SBIR project (5R44AG074116-03). During this period, we have accomplished the following milestones that will serve as a foundation and preliminary work for the development of the Olera CareNavigator in Phase IIB:

- Attracted an online community of over 10,000 family caregivers and older adults looking for care support, many interested in participating in research studies with Olera, Inc.
- Conducted thematic analysis on AD/ADRD caregiver interviews to identify prominent challenges, defining crucial requirements for digital care planning tech.[5]
- Established a nationwide database called Educational Materials and Caregiver Resources (EMCR), featuring 30,000+ resources including provider profiles, support groups, and more.
- Created a user-friendly Web-based Care Planning Tool with novel Dementia Care Personalization Algorithm to match needs and local service providers and education for family caregivers, scoring 4.6/5 on the 'Mobile Application Rating Scale' among 25 caregivers, indicating excellent usability.[24]
- Developed an initial LLM senior care planning prototype to demonstrate the preliminary feasibility of fine-tuning a pre-trained LLM to be elder-care domain-specific laying groundwork for Phase IIB innovation.

Ensuring Safety, Privacy, and Accuracy for Users while Mitigating Risk of AI

Our team is committed to safety, privacy, and accuracy in deploying LLMs for senior care planning. Using RLHF, we collaborate with experts to refine technology development, with a thorough focus on anonymizing and securing private data. Prior to a public release, we rigorously test in-house for optimal performance and conduct third-party audits to ensure adherence to safety, privacy, and accuracy standards.

INNOVATION

Key Innovation 1: Eldercare-specific Large Language Model (LLM) for Increased Care Accessibility and Affordability. LLMs, a specific category of machine learning algorithms adept at processing natural human language, offer a unique ability to numerically represent and process language for computational and algorithmic tasks. This capability is particularly valuable in navigating the eldercare ecosystem where individuals (1) may not be familiar with eldercare terminology, (2) use different terms to describe the same things, or (3) are uncertain about their specific needs. With the improved capability to dynamically handle natural language from users, LLMs can significantly reduce the need for boilerplate information, surpassing static, one-size-fits-all information delivery and ushering in a new era of adaptive, context-specific output in digital care planning technology. However, it's essential to underscore that without proper domain-specific training, these LLMs can yield ineffective or inaccurate outputs or responses to prompts.

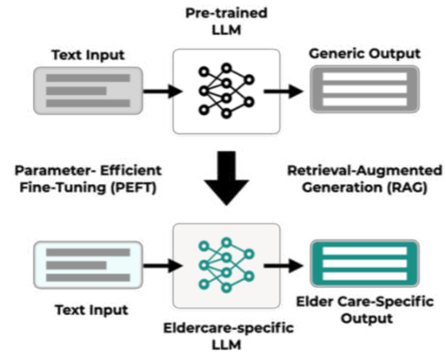

Figure 3: Illustrative overview of the Eldercare-specific LLM

To address this challenge, an innovative eldercare-specific LLM will be created by fine-tuning an existing LLM called a generative pre-trained transformer (GPT) taking advantage of two recent innovations in LLM training – Retrieval-Augmented Generation (RAG)[25] and Parameter-Efficient Fine-Tuning (PEFT).[26] By implementing RAG, the elder care tool can dynamically access a vast databases of caregiver resources (housed in our EMCR database). This enables the tool to provide comprehensive guidance without expending large costs by training a model from the ground up. Meanwhile, PEFT allows us to adapt pre-existing models like GPT to the specific requirements of the elder care domain, especially the ability to perform key care planning functions like needs assessment, resource matching, and personalized instruction delivery. By fine-tuning only a subset of the model's parameters, PEFT makes the process more resource-efficient and accessible. Combining RAG's informed retrieval with PEFT's customization creates a powerful solution to approaching training an LLM to be capable of care planning functionality (Figure 3).

Key Innovation 2: A Suite of Independent but Collaborative Agents for Improved Elder Care Planning
Elder care planning involves assessing needs, estimating costs, finding financial or other aid programs, and devising personalized and clear guidance on how to pay for needed care and access available support systems. To address this, we propose a suite of AI agents acting as virtual assistants to navigate complex and robust data on caregiving resources and provide personalized guidance to improve access to existing aid and support systems. Our multi-agent system features specialized AI agents handling specific care planning functionalities while collaborating to leverage expertise. Ultimately, these agents will combine into a Multi-Agent Network capable of transforming raw input into on-demand guidance personalized to a caregiver's unique circumstances and responsive to caregiver-specific inquiries.

Our multi-agent platform features three types of agents: **(1) Needs Assessment Agents** with novel elder care-specific LLM to effectively evaluate gaps in care and challenges or needs; **(2) Resources and Aid Gathering Agents** capable of resource-matching, eligibility-screening, and sifting through thousands of available support options, aid programs, and other resources housed within the Educational Materials and Caregiver Resources (EMCR) repository developed by Olera, Inc. The EMCR repository is diverse, including information spanning transitional and non-traditional healthcare services including community and government programs, nonprofits and charities, community support services, and veteran services among others; and **(3) Personalized Care Planning Agents** that explore and rank suitable resources and provide instructional guidance output delivered to the user and to be optimized using ML through reinforcement learning from human feedback (RLHF) with domain experts in licensed clinical social work (LCSW). Agents will then be integrated into a Multi-Agent Network to perform holistic care planning functions. Figure 5 puts the CareNavigator at the center to showcase the AI agents to learn from one another providing holistic guidance to be delivered by the CareNavigator.

RLHF will be used to integrate human domain expert training and eventually direct user input for informed adaptation of the system and more accurate responses over time.[25] LCSWs will serve on an RLHF panel to continuously optimize the AI agents' functionality, and training, and contribute to training data included in the EMCR to ensure responses meet best-practice standards in information accuracy, quality, and delivery. A coded reward system will be built into the system to further train the system in an ongoing process utilizing user reactions to generated responses and encounters with the CareNavigator. For example, individual users will be capable of rating responses generated by the system, indicating relevance or usefulness, among others.

Key Innovation 3: Inclusive Virtual Elder Care Conversational Agents as an Intelligent User Interface (UI) for Multi-Agent Network. Recent advances in generative AI enable the creation of inclusive personalized avatars capable of conversational human-like conversation and offer the opportunity to increase the usability of technology being shown to be particularly promising in performing better than traditional user interfaces in older adult users.[25,27] We plan to synthesize realistic avatars reflecting diverse ages, ethnicities, and backgrounds to provide customized elder care guidance through enhanced relatability. This novel approach aims to increase engagement by leveraging relatable avatars and a human-like conversational encounter to provide individualized guidance for caregivers (Figure 4).

Figure 4: Sample avatar agents for the Olera CareNavigator created using generative AI

Final End-Product: The Olera CareNavigator. (Figure 5) This AI-powered care planning tool aims to provide personalized responses that ultimately simplify complex data into tailored guidance for family caregivers to better problem-solve, qualify for aid, and ultimately access affordable care solutions. In practice, the key value lies in its ability to provide clear and efficacious responses to a user on how to access available support systems and deliver information that leads to positive behavior change. To our knowledge, there are no existing solutions currently available that leverage these recent advances in AI technologies to provide personalized guidance on the affordability and accessibility of care for the elderly population.

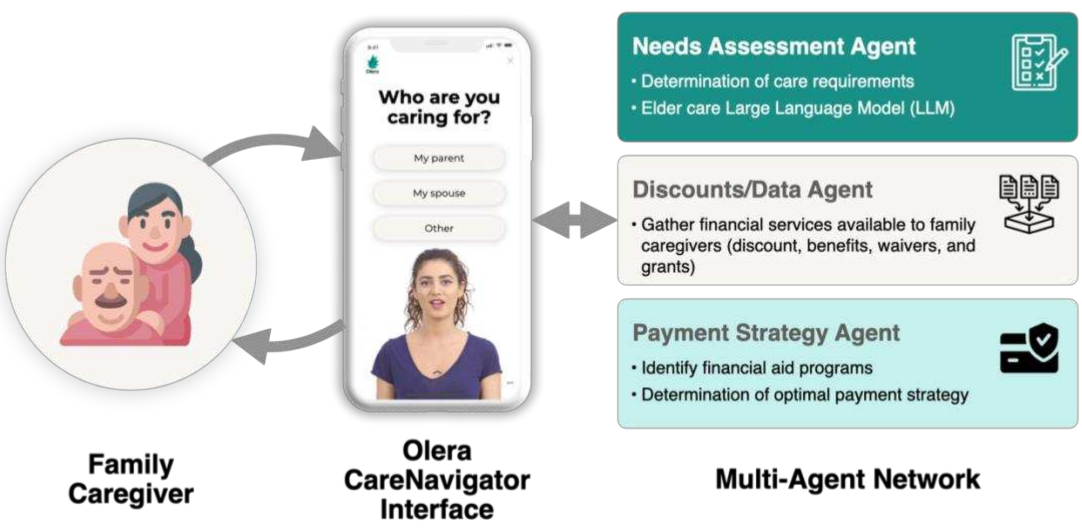

Figure 5: An illustration of the user journey and how it interfaces with the Olera CareNavigator working with the Multi-Agent Network

An Interdisciplinary & Innovative Team. Our team is experienced in artificial intelligence research, aging research, clinical medicine, and software development making us uniquely suited to execute this project. The team is Led by Olera, Inc. CEO, Tokunbo "TJ" Falohun, a former Pfizer engineer and serial med-tech entrepreneur with a patent for the use of artificial intelligence technologies in ocular disease diagnosis. Dr. Logan DuBose, MD, MBA, the Chief Operating Officer of the company will serve as a co-investigator and provide clinical expertise relating to people living with dementia. He will assist Mr. Falouhn in directing required activities to meet milestones. Dr. Jim Nolan, Ph.D. is an expert on Large Language Models, deep learning, and transformers, will lead the development of the multi-agent network. He has previously worked with NIAID and NIH on LLMs to detect misinformation about vaccines and infectious diseases showing his deep commitment to applying AI to healthcare concerns. Human subjects research will be supervised by Dr. Marcia Ory, Ph.D., MPH, MS, the founding Director of the Center for Population Health & Aging at Texas A&M University with over 20 years in federal service as chief of Social Science Research on Aging in the Behavioral and Social Research Program at the NIA. Dr. Qiping Fan, DrPH, MS, Assistant Professor of Epidemiology at Clemson University with over 6 years of experience in conducting funded public health research, will be leading human subjects research efforts with Principles and Dr. Ory in areas such as study design, data analyses, and scientific reporting with further guidance from senior biostatistician Gang Han, PhD., Professor of Epidemiology and Biostatistics, Texas A&M University. This leadership team is further supported by diverse interdisciplinary colleagues spanning community partners, industry providers, private equity groups, and academia as outlined in the budget justification and letters of support.

PRELIMINARY DATA

Investigation of Top Caregiver Needs & Preliminary Evaluation of Web-based Care Planning Solution**.** Through qualitative analysis and 200+ caregiver interviews in Phases 1-2, we identified unmet needs in financial, legal, and care planning domains and caregiver expectations for care planning tech. Equipped with insights from real caregivers, we developed a personalized web app matching caregivers to tailored educational materials and care services. End-products from Fast Track SBIR efforts include:

- Developed and evaluated usability (n=31) of a novel web-based care planning tool for AD/ADRD family caregivers that achieved a 4.6/5 on the MARS scale for digital health applications[24] (Figure 6)
- Developed a growing nationwide database of over 30,000 resources in the EMCR database for caregivers and older adults to access via a web-based care planning tool[28]

Figure 6: Caregiver support platform

Preliminary Prototype of Senior Care Focused LLM**.** Preliminary work in previous SBIR work has demonstrated that LLMs when trained and augmented correctly, can provide

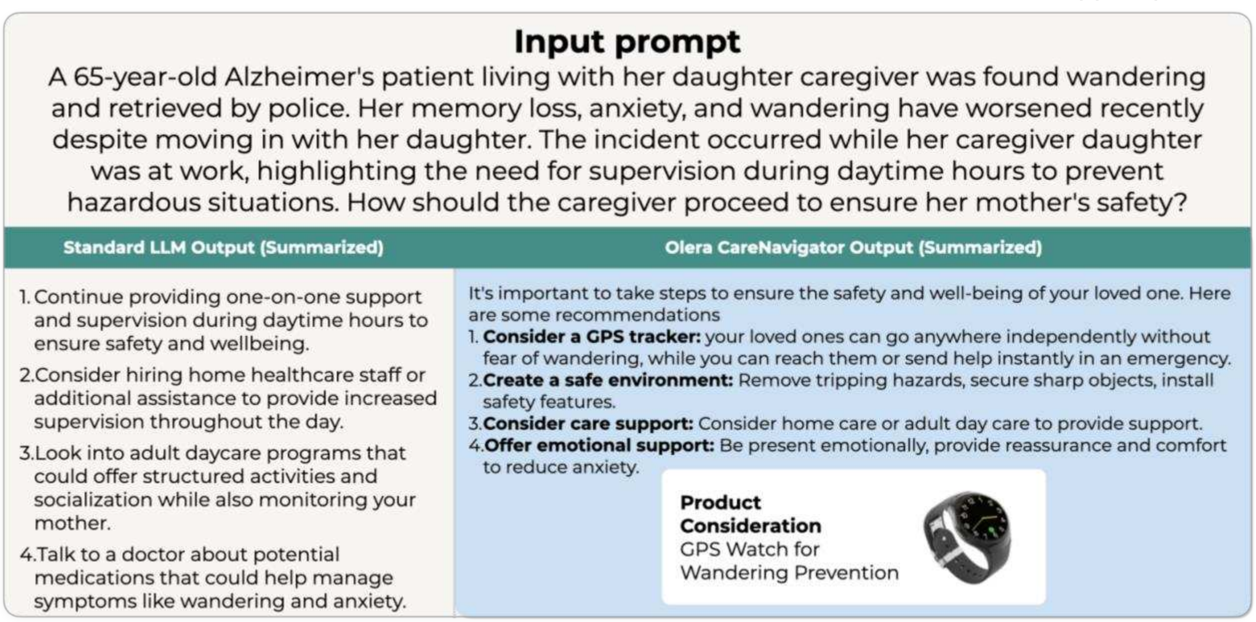

Figure 7: Demonstration of the early functionality of the custom LLM prototype. The input prompt was inserted into a standard LLM as well as the custom LLM. The results have been shortened for presentation purposes

personalized information and instructions relevant to a unique caregiver situation. This shows the potential to assist families in navigating the complex elder care ecosystem in multiple domains to secure access to aid programs and find affordable local care and support solutions. Figure 7 shows a prototype eldercare-specific LLM developed in preliminary work leading to this project with the capability to improve responses based on fine-tuning of a pre-trained GPT to demonstrate the feasibility of improving personalized guidance and resource matching using AI.

APPROACH

Specific Aim 1: Development of Specialized Agents for Elder Care Planning

Rationale: Our approach breaks down a process of senior care planning into manageable parts for greater holistic and personalized responses for caregivers (Figure 8). Specialized AI agents focus on key tasks - needs assessment, resource & aid matching, and personalized care planning to identify needs, perform resource-matching, and present personalized information on how to access or utilize relevant support systems. Figure 8 shows an overview of the development process of the specialized agents in Aim 1.

Figure 8: Overview of the development process of the specialized agents

Task 1.1: Development of the Needs Assessment Agent. Under this task we will develop 1) agents that discern specific care requirements for seniors and caregivers based on provided text or verbal information input and 2) an eldercare-specific Large Language Model that generates accurate responses to caregiver inputs and inquiries regarding elder care resources, services, or general care-related problems or concerns.

(Task 1.1A) Develop an elder-focused Focused Large Language Model: We will develop an elder care GPT LLM using Retrieval-Augmented Generation (RAG)[25] and Parameter-Efficient Fine-Tuning (PEFT).[26] RAG allows dynamic access and information retrieval from external databases without the expense of full model training. Leveraging GPT4All and Llama 2, we will enhance the model with elder care data to specialize it without custom costs typically required for development. PEFT enables efficient customization by modifying key model parameters, resulting in specialized models for daily activities, medication, and emergencies among others. Combined with RAG, PEFT provides a means to create an efficiently tailored LLM specialized that is responsive to unique care challenges and capable of information retrieval from external databases with information on caregiving support programs, aid, and care services.

(Task 1.1B) Identification of Need Parameters: In this task, we will identify parameters that define elder care needs. These will include the factors like presence of pre-existing health conditions, functional abilities (ability to perform daily activities independently, cognitive status, social support, mental well-being, and mobility. Once identified, these parameters serve as the systems information processing infostructure to enable understanding and predicting the functional needs of an individual user.

Metrics for Success for the Needs Assessment Agent (Task 1.1):

- **Information retrieval accuracy** using precision, recall, and F1-score (>90% target)
- **Response quality** via BLEU (>0.7), ROUGE (>0.5), and expert ratings (>80% target)

Task 1.2 Development of the Resource & Aid Matching Agents. To sift through thousands of data points in the EMCR database (and also other partnering external data repositories with care support resources) and identify suitable resources and aid, we will develop two subagents that constitute the Resource & Aid Matching Agents: 1) a Data Gathering Subagent and 2) an Eligibility Screening Subagent, both of which are detailed below.

(Task 1.2.A) Development of the Data Gathering Subagent: This subagent will gather data from the EMCR database and partnering information repositories like eldercare.acl.gov, Medicare.gov, Alzheimer's Association, ClinicalTrials.gov, local Aging and Disability Resource Centers (ADRC) stations, and Census.gov among others. This data on aging, local services, costs, research, demographics, and trials can be quickly queried and aggregated for the eligibility screening subagent discussed below.

(Task 1.2B) Development of Eligibility Screening Subagent: This subagent matches user needs, circumstances, and preferences to the most suitable aid programs or support services in the aggregated database created by the Data Gathering Subagent. When users provide details such as demographics, financial status, and care needs, an algorithm sifts through data comparing those details with eligibility criteria and determining the likelihood of qualifying. A 'suitability score' will be given to each resource or aid program based on how well the criteria match the user's details. The more criteria match, the higher the program scores. By sorting programs according to their scores, we ensure that the information presented to the users is relevant and tailored to their needs, preferences, and circumstances.

Metrics for Success for the Resource & Aid Matching Agents (Task 1.2):

- **Data Accuracy**: Comparison of scraped data against manually verified data points to ensure representation accuracy (>90% match when comparing scraped data to verified data)
- **Completeness**: Verification of information extracted without missing fields or partial data (>90% field completion rate for all required data fields).
- **Efficiency**: Optimization of the scraping process to minimize time while maintaining accuracy and completeness (target average scraping time <5 seconds/page while maintaining above 90% accuracy and completeness)
- **Eligibility-Matching**: Comparison of generated suitability scores against manually calculated scores to evaluate resource-matching accuracy (>90% accuracy when comparing generated vs. manually calculated scores).

Task 1.3: Development of Personalized Care Planning Agents. Once the previous agents process their inputs, the Personalized Care Planning Agent consolidates information and communicates tailored information to the user in a clear, straightforward manner (free from technical jargon or complex terminology) and formatted to resemble instructions or “care plans” with a focus on expense management and resource access. To help families better understand the possible care-related expenses, an optimal payment strategy subagent will project out-of-pocket costs for any needed support and devise an effective payment strategy based on available finance options. Further, our elder care LLM delivers final detailed instructions to access support systems including information like application steps, required documentation, and support phone numbers.

(Task 1.3A) Development of an Optimal Payment Strategy Agent: Devising an effective payment strategy for elder care requires a customized approach based on the individual's financial situation and care needs. We will build a financial model incorporating individual budget limits, asset preferences, aid program eligibility, and other constraints like insurance coverage. Optimization techniques like reinforcement learning, neural networks, decision trees, constraint satisfaction, and time-series analysis will enable personalized strategies with an ideal breakdown of funding sources, balancing cost, quality, and individual preference or circumstance.

(Task 1.3B) Communication of Personalized Care Plans to Caregiver in Natural Language: The output of the previous agent will be an optimized financial strategy represented in a mathematical, graphical, or some other complex language unfamiliar to most caregivers. We will use our novel eldercare-specific LLM (Task 1.1) to convert it into a language that is easy to understand for all caregivers, regardless of their financial literacy or familiarity with common elder care jargon. We will also fine-tune the LLM to provide the payment plan in multiple languages for users with diverse backgrounds.

(Task 1.3C) Determination of Optimal Care Plans through Reinforcement Learning from Human Feedback: To address potential errors, we will use Reinforcement Learning from Human Feedback (RLHF). Domain experts who are licensed clinical social workers (LCSW) will evaluate all information and care plans delivered and provide feedback on any suboptimal information (see letters of support for RLHF panel members). This will be incorporated as a reinforcement signal to iteratively improve the model's personalized care planning abilities and align them with expert knowledge.

Metrics for Success for the Payment Plan Agents (Task 1.3):

- **Optimization success** via cost efficiency, feasibility, solution quality, and alignment with caregiving objectives. Specific metrics will be a cost reduction of at least 20% in proposed simulated use cases
- **Natural language conversion** through readability, coherence, accuracy in representing the strategy, and personalization. Techniques include Flesch-Kincaid, discourse analysis, semantic similarity, and domain expert review. Specific metrics will be a Flesch-Kincaid grade level >6 for readability, discourse analysis coherence score >80%, and a semantic similarity >0.7 when compared to ideal messages.
- **RLHF performance** via adaptation speed, user satisfaction, behavioral alignment, and feedback utilization. Specific metrics will be adaptation speed <5 seconds, domain expert satisfaction scores >4 on a 5-point Likert Scale, behavioral alignment with caregiving objectives >90%, and a feedback utilization rate > 85%.

Specific Aim 2: Integration of Multi-Agent Network & Preliminary Evaluation of a Novel Intelligent User Interface (UI) Among AD/ADRD Caregivers.

Rationale: The agents will work together seamlessly in a Multi-Agent Network with each agent providing data to the others. For example, the needs assessment agent will provide information on functional requirements to the optimal payment strategy agent, which will be used to formulate financial strategies for required care. Simultaneously, the resource & aid matching agents will identify potential financial aid programs to help save costs, which can then be incorporated into the financial strategies. The system will be designed to learn and evolve, and the modular architecture allows for continuous updates and improvements in one agent without disrupting the others. Specifically, a Microservices Architecture where agents are independent, modular services communicating via APIs. This enables separate deployment, scaling, and optimization per service. Decoupled services increase resilience by isolating issues. Containerization and orchestration allow smooth management and deployment.

Task 2.1: Integration of Multi-agent network. Creating the proposed Multi-Agent Network requires the development of service-specific APIs to allow efficient interaction among agents and the implementation of data exchange standards. Our agents will be exposed as RESTful services (generally highly scalable, light, and maintainable APIs for web-based applications) that can be accessed and composed over the network. To achieve this, we will implement the services using FastAPI. FastAPI has been chosen as the primary tool to expose the services of the Olera CareNavigator system due to its high performance, automatic documentation generation, and robust input validation through Python's type hints. Its support for both asynchronous and synchronous code, coupled with security features and ease of development, ensures a scalable and efficient API interface. By utilizing FastAPI, the system provides a seamless, secure, and user-friendly interaction between the agents. Communication between the agents will be standardized using JSON data structures.

Potential pitfalls and alternative strategies. The complexity of a microservices architecture in the context of specialized planning agents poses a potential pitfall. Managing numerous services can introduce complications in deployment, communication, data consistency, and fault isolation. To overcome this, comprehensive monitoring and logging systems can be employed using tools like Prometheus and Grafana, which allow for real-time analysis of performance metrics. Implementing container orchestration through Kubernetes assists in automating deployment, scaling, and management.

Task 2.2: Development of Intelligent User Interface (UI) Avatar. The UI is critical for caregivers to adopt our AI-powered system. Rather than inanimate chat boxes, we use custom-made and culturally diverse life-like avatars to make the UI more accessible and dynamic for caregivers. The AI Avatar will be a virtual conversational agent that uses Natural Language Processing (NLP) to replicate human interactions. We will leverage the use of generative AI frameworks from Synthesia.io and ElevenLabels (audio generation) to construct the avatars. Our approach is supported by research from Ponathil et al. (2020) and Wang (2022), who found that avatars are superior to traditional UIs in terms of workload and usability.[27,29]

Task 2.3: Human Subjects Pilot Testing the Usability and Functionality of Integrated Multi-Agent Network and User Interface. A mixed-method evaluation will test the usability and functionality of the developed technology. The quantitative assessment of the Olera CareNavigator Digital Platform will be conducted using the survey guided by the framework of User Experience Evaluation in Intelligent Environments (UXIE),[30] a comprehensive framework to assess the user experience in intelligent environments through in-depth individual interviews collecting qualitative data on aspects of acceptance, usability, adaptability, appeal and emotions, intuitiveness, unobtrusiveness, safety, and privacy. The UXIE framework considers user experiences across the technology lifecycle - before, during, shortly after, and long after use. The explanatory sequential mixed-method design enables collecting rich feedback on the platform's features, information quality, and overall value to identify enhancement opportunities.

Statistical Design and Power. The null hypothesis for the quantitative assessment is that there is no difference in mean scores of the UXIE survey before and after using the Olera CareNavigator Digital Platform. To detect a medium (d = 0.5) effect size, the least sample size is 15 to compare before and after differences achieving 80% power at a significance level 0.05.[31] Considering the common retention rates of caregivers of PLwD (35% to 38%) indicated by previous research,[32] we plan to enroll at least 25 caregivers of PLwD. The selection of the pilot group will consider the demographic distribution of the caregivers of persons living with dementia (PLwD) in the 2023 Alzheimer's Disease Facts and Figures[33] and the caregiving population distribution in the Texas area.[34] The technology use and comfort level will be assessed using validated questions such as items from the Technology Proficiency Self-Assessment (TPSA) questionnaire,[35] which has been widely used as a technology self-efficacy measure. We will assess the social needs of caregivers using the Health Leads Tool,[36,37] which evaluates critical social needs domains such as food insecurity, housing instability, utility needs, childcare difficulties, financial resource strain, transportation challenges, literacy, and lack of social support. For the complementary qualitative study, data saturation[38] will be used to determine the sufficiency of sample size and end of data collection.

Data Collection & Analysis. We will identify a pilot group of caregivers of PLwD (N=25) recruited from our Healthy Aging Community (+10,000 members), aging and AD-related organizational partners in Texas (see letters of support) and selected from our current user database to roughly represent the diverse range of our end-users (e.g., both male and females, spouse and child caregivers; at least 30% racial or ethnic minorities). The chosen pilot group will be given a formal introduction to the system with a focus on its purpose, capabilities, and uses, followed by a comprehensive training session to ensure users are comfortable and adept at using the system. Allowing them to interact with the CareNavigator independently is key for in-depth observations and discussions.

The participant's group will then use the system over a predetermined period (around four weeks), with periodic routine checks scheduled for data collection, such as UXIE surveys, system usage metrics, and feedback sessions through interviews. Observations around acceptance, usability, adaptivity, appeal, and emotions — particular aspects that work well, and those that pose difficulties—will be crucial at this stage. The data triangulation through UXIE surveys, interview feedback, and usage metrics observation will be analyzed to understand the user experiences of caregivers interacting with the CareNavigator. The findings will inform an iterative product design process using the Build-Measure-Learn framework[39] where we will make necessary refinements to the system to increase user satisfaction, improve functionality, adaptivity, appeal, emotions, and streamline usability before larger scale testing in Aim 3.

Milestones/Criteria of Success

- **Seamless Inter-Service Communication:** APIs enable seamless agent interactions (95% success)
- **Responsiveness:** Real-time updates within 1 sec (99% met)
- **User Experience**: Quantitative assessment by UXIE surveys (90% positive responses in Likert questionnaire) and qualitative feedback by interviews (90% caregivers providing generally positive feedback)
- **Qualitative Research Quality**: Consolidated criteria for Reporting Qualitative research (COREQ) Checklist (100% reported and 90% met)

Specific Aim 3: Evaluation of the User Experience and Caregiver Perceptions of Novel AI-Powered Care Planning Technology Among AD/ADRD Caregivers with Diverse Backgrounds.

Rationale: This evaluation will include the acceptability and usability of the platform and caregiving experience among human users from diverse backgrounds and settings to obtain a more comprehensive view of the caregiver experience interacting with the technology. In a cohort of caregivers of PLWD (n=200), we will assess these outcomes after use of the end-product of Aim 2 using the following validated tools: The Mobile Application Rating Scale (MARS),[40] modified Technology Acceptance Survey (TAS),[41] the Positive and Negative Appraisals of Caregiving (PANAC) and the Caregiver Self-efficacy scale (CSES-8).[42]

Research Methodology

Overall Framework. To measure caregivers' perceptions, we will use the PANAC and the CSES-8 survey instruments.[42,43] The PANAC is a concise yet comprehensive instrument to assess the holistic caregiving experience concerning individuals providing care for those with dementia. It offers a streamlined approach to evaluate various caregiving domains, such as the emotional, physical, and financial burden of caregiving, as well as the positive aspects of caregiving, such as a sense of purpose and satisfaction.[42] The CSES-8 is a short and comprehensive tool to examine caregiving self-efficacy levels (the confidence caregivers have in managing caregiving responsibilities and challenges). It provides insights into effective interventions by linking greater self-efficacy to lower distress as more confident caregivers tend to experience less burden navigating their roles.[43] We will also collect ethnographic data on social and individual factors to examine how technology experience and caregiving differ among caregivers with varying levels of social needs The Health Leads Tool will be used to assess social needs like food, housing, childcare, finances, transportation, literacy, and social support. Further, we will also roughly stage PLwD cognitive functioning by using the Quick Dementia Rating System (QDRS)[44] to be used as a covariate in our analysis.

Selection and Recruitment of Diverse Participant Cohorts. A diverse socioeconomic cohort of dementia caregivers will be recruited from Texas. We chose Texas for recruitment as our company was launched there, and it offers a large diverse population of caregivers, aligning with our study's goals. We will plan to recruit at least 50% of the subjects from individuals indicating at least one domain of social needs from the Health Leads tool (e.g., food insecurity, housing instability, utility needs, child care difficulties, financial resource strain, transportation challenges, literacy difficulties, lack of social support). We plan recruitment through local channels that interface with low-socioeconomic status households and closely monitoring baseline survey responses of enrolled participants (reported household income and perceived social needs in survey responses). In addition, we will recruit through Alzheimer's Association chapters and Area Agencies on Aging caregiver networks in Texas. Participants must meet the following eligibility criteria for study enrollment: (1) be the primary non-paid caregiver of a person living with dementia (PLWD) (including a mix of mild, moderate, or severe cases) in Texas, (2) provide at least 10 weekly hours of care for a PLDW who is yet to be institutionalized, (3) be the adult child spouse, or family member of the PLWD, (4) have a concern about or perceive a need for more information on financial management and/or legal planning for caregiving, (5) have access to a smartphone or computer with internet access. Formal (paid) caregivers will be excluded from this study.

Evaluation of Integrated Network Among Diverse Populations. We will conduct an evaluation of the acceptance and perceived value of the specialized agent suite across 200 participants. Participants will complete an enrollment survey assessing demographic characteristics, socioeconomic status, technology comfort level, caregiving contexts (e.g., weekly hours devoted to caregiving, caregiving experience, and self-efficacy), and caregiving resources (e.g., any social needs such as food insecurity, financial strains), as well as their PLWD's sociodemographic and health characteristics (e.g., dementia stage as indicated by Quick Dementia Rating Scale). This will provide statistical data regarding the demographic distribution of the caregivers and their initial perceptions of the specialized agent suite.

<u>Participant Training, Engagement & Ethical Considerations.</u> Upon completing the enrollment survey, participants will be guided to the Olera CareNavigator to use. While participants engage with the CareNavigator avatar, the research team will monitor and record data including task completion status, completion time, and other interaction metrics. Observational notes will be taken to capture important insights. After interacting with the agents, participants will be prompted to complete the Mobile Application Rating Scale (MARS) and Technology Acceptance Survey (TAS) that incorporate platform features and functions such as assisting with financial planning. Next, participants will be asked to engage with the Olera CareNavigator for 4 weeks, during which both user experience data from the platform (such as frequency of access, and types of activities engaged) will be collected. At least one technical support call will be provided to each participant during this period to ensure smooth utilization and to gather feedback. At the end of the engagement activities, participants will be invited to complete MARS and TAS surveys to assess their acceptance and perceived usability of the technology after 4 weeks of interacting with it. After 3 additional months of using the technology, participants will be invited to complete the CSES-8 and PANAC to assess their caregiving experience using the technology. Approval will be obtained from the Institutional Review Board at Clemson University and all participants will be asked for informed consent before beginning the study. The study's sensitive nature requires special attention to confidentiality, cultural sensitivity, and respect for individual autonomy. Appropriate measures will be taken to ensure that these principles are adhered to throughout the research process.

<u>Outcome Measures, Data Analysis, and Product Iteration.</u> The primary outcome measures for this study are technology acceptance (measured by TAS), technology satisfaction (measured by MARS), and caregiving confidence and experience (measured by CSES-8 and PANAC). The secondary outcome measures are the sub-constructs of MARS (e.g., engagement, functionality, aesthetics, information, subjective quality) and TAS (e.g., usefulness, perceived ease of use). Descriptive statistics (mean and standard deviation or frequency and percentage) will be used to summarize the characteristics of caregivers and their engagement experience with the Olera CareNavigator. The multivariable regression models will examine changes in primary outcomes before and after caregivers' engagement with the technology, adjusting for socio-demographic characteristics and social needs factors. Additionally, open-ended responses and observational notes will be coded and categorized into themes to provide supplemental insights into caregivers' experiences and perceptions.

<u>Power calculations.</u> The null hypothesis is that there would be no differences in the overall technology acceptance and satisfaction scores, and caregiving experience indicators (CSES-8 scores and Positive

Appraisal/Negative Appraisal Ratio) before and after using the technology, and between caregivers with zero or one versus multiple aspects of social needs. We aim to detect the specific effects by the difference in mean of the overall acceptance and satisfaction ratings in the TAS and MARS questionnaires among participants with 0–1 and >1 aspects of social needs, and the mean of caregiving self-efficacy rating in the CSES-8 questionnaire and the Positive Appraisal/Negative Appraisal (PA/NA) ratio among participants before and after using the technology for 3 months. To detect large (d = 0.8), medium (d = 0.5), and small (d = 0.2) effect sizes based on benchmarks suggested by Cohen[31,45] and achieving 80% power at significance level 0.05, the least sample size for each group will be 25, 63, and 392 respectively to compare between-group differences for the first hypothesis, and the least sample size for all participants will be 6, 15, and 96 respectively to compare before and after differences. Depending on the level of effect sizes, the required sample size ranged from about 50 to 794. Past research has indicated low retention rates for Alzheimer's disease research, ranging from 35% and 38%.[32] Following our experience in recruitment and retention in similar projects at the Texas A&M Center for Community Health and Aging, we estimate that a minimum total sample size of 126 for the conclusion surveys and 200 for the enrollment surveys will be required. This sample size is both practical and sufficient to detect a medium effect size of 0.5 for between-group differences, and a small effect size of 0.2 for before-after differences.

Milestones/Criteria for Success

- **Technology Acceptance**: At least 4.0 on the TAS scale (ranging from 1 to 6)
- **Technology Satisfactio**n: At least 3.6 on the MARS scale (ranging from 1 to 5)
- **Caregiving Self-efficacy**: Increasing CSES-8 scores after using the technology compared to their scores before usage (>10% increase)
- **Caregiving Experience**: Higher PA/NA ratio after using the technology compared to their PA/NA Ratio before usage (>10% improvement).[42]

Potential Problems and Alternative Approaches. Given the diversity of dementia caregivers we are targeting, a possible challenge is recruiting enough participants with diverse socioeconomic characteristics, as participant recruitment is a well-known challenge in dementia intervention research. However, by working closely with the Alzheimer's Association, and leveraging our online community of 10,000+ family caregivers, we believe we would be able to reach a large number of dementia caregivers.

TIMELINE

Approximate study timeline for major study activities.												
Year	**1**				**2**				**3**			
Quarter	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Protocol, IRB, and Training												
Protocol development	X	X	X	X	X	X						
IRB approval/amendment		X	X	X	X	X						
Staff hiring/training	X			X	X							
Development of User Interface												
Agent development	X	X	X									
Agent refinement		X	X	X								
Agent integration				X								
Study 1: Preliminary evaluation												
Recruitment			X	X	X	X	X					
Data collection				X	X	X	X					
Data analysis							X	X	X			
Study 2: Comprehensive evaluation												
Recruitment					X	X	X	X	X	X	X	
Data collection						X	X	X	X	X	X	X
Data analysis											X	X
Dissemination and reports												
Quarterly reports	X	X	X	X	X	X	X	X	X	X	X	X
Study product dissemination				X	X	X	X	X	X	X	X	X

Commercialization Plan

A) Problem, Solution, and Value of the SBIR Project

Problem. America faces a caregiving crisis. For many families, senior care is unaffordable,[1] yet millions of eligible seniors miss out on over $30 billion in unclaimed aid that could assist with essential needs like food, housing, prescriptions, and healthcare.[2] This is largely due to the difficulty in identifying and enrolling in relevant support programs or initiating appropriate care services that apply to local senior care resources.[2] This challenge is heightened for people with Alzheimer's disease and related dementias (AD/ADRD), where the resources tend to be fewer and more expensive.[3] In 2023 alone, AD/ADRD will cost America $345 billion, not including the 18 billion hours of care provided by unpaid family caregivers valued at $339.5 billion.[4]

It is clear family caregiving is the backbone of AD/ADRD care in the US, and this comes at a cost. Over 11 million US AD/ADRD caregivers face significant financial, psychological, and physical challenges –collectively called caregiver burden– with prevalence as high as 43% reported.[3] Dementia care's lifetime cost is about $392,874, with 70% covered by families through out-of-pocket expenses and unpaid care.[5] Caregivers also face increased risks of depression, anxiety, social isolation,[6] and physical health issues like chronic stress and hypertension.[7] As worldwide AD/ADRD cases could triple by 2050,[8] addressing the caregiver burden becomes a significant public health concern demanding urgent innovation.[9] In response, our prior Fast-Track SBIR project investigated the specific challenges contributing to caregiver burden.[10] Working with over +200 AD/ADRD family caregivers, we identified two prominent challenges in AD/ADRD caregiving: (1) high costs of care and (2) under-utilization of resource, aid programs and care services.

Challenge 1: High Cost of Care. Simply put, senior care, particularly for those with AD/ADRD, is not affordable for most. This significantly affects the well-being of individuals and their families.[11] According to the Genworth Survey of Long-Term Care, the average cost of memory care in the US is $114,007 annually full-time home care costs $59,540, and 7 out of 10 people will require long-term care like this in their lifetime.[12,13] With the average American lacking retirement savings and 1 in 3 older adults classified as "economically insecure" with incomes below ~$50,000, relying on public assistance like Supplemental Security Income,[14] Social Security Disability Insurance (SSDI),[15] and Medicaid is common.[16] However, these aids fall short of covering all dementia care expenses, and caregivers face, on average, $280,000 in out-of-pocket costs for unpaid care throughout their caregiver journey.[5] Consequently, around 40% of AD/ADRD caregivers quit jobs due to caregiving demands, exacerbating financial stress through lost wages, job instability, and depleted savings.[17]

Challenge 2: Under-utilization of Resources, Aid and Care Services. Strikingly, older Americans leave $30 billion in unclaimed financial resources for necessities like food, prescriptions, utilities, and healthcare.[1] For instance, almost half of eligible seniors miss out on the $6.3 billion annual aid from the Supplemental Nutrition Assistance Program (SNAP) due to unawareness or confusion about enrollment.[2] Complex application processes and public confusion about residential and long-term care systems can hinder resource utilization.[2] For functional care assistance, initiating services like home health aid or senior living communities involves time-consuming tasks such as calls, interviews, internet research, and paperwork.[10] Language barriers, literacy, and technology limitations could make tasks like these even more complicated and risk leaving many without vital support.[18]

Personalized Care Planning. Practical solutions to these challenges come from licensed and highly trained clinical social workers (LCSW) and experienced case managers offering personalized holistic care planning for family caregivers.[19] They specialize in resource optimization and care coordination, acting as advocates and advisors to secure care for individuals with financial, medical, or mental health challenges. This form of personalized and tailored assistance is the more effective delivery of caregiving resources and education.[20] However, shortages, cost, and high demand limit their accessibility,[21] possibly explaining why ongoing dementia caregiver support from social workers is not widespread despite its potential to reduce utilization of healthcare services and enhance patient experiences with care.[22] This sets the stage for impactful innovation. Digital and AI advancements could capture some value social workers provide and scale it through widely accessible online care planning technology. Building on this, our prior SBIR project explored caregivers' desires for web-based care planning tools.[10] They expressed interest in features like:

- Customized support and resources for emotional relief and care planning
- Financial guidance for cost-effective solutions, benefits, and community aid
- Comprehensive service database for caregiver resources like aid programs and care services

Solution. In response to these findings, we're developing the Olera CareNavigator—an AI-powered care planner engineered to make senior care more affordable and accessible (Figure 1). It offers general support for AD/ADRD and other senior care challenges with specific functionalities to help 1) find affordable care solutions to meet needs and 2) better utilize resources, services, and aid programs. Our approach involves using Large Language Models (LLMs) to create three "AI agents" that specialize in tasks related to senior care planning, such as assessing needs, identifying relevant local support resources, and providing personalized responses to caregiver questions. To ensure comprehensiveness in recommendations, it integrates with a robust caregiver resource database with information on aid programs and care services on the local, state, and federal levels developed in previous SBIR work by Olera, Inc. (EMCR database). To make it adoptable by caregivers from diverse backgrounds, it will be hosted on an intelligent user interface (UI) and web-based dashboard with an AI Avatar capable of human-like conversation, creating a user experience similar to an initial or follow-up encounter with a professional social worker — functionality seemingly impossible before recent advances in generative AI.[23,24] Quality will be maintained through fine-tuning the Multi-Agent Network with domain expert LCSW through a machine learning technique called reinforcement learning from human feedback (RLHF).[25] For a caregiver-centered user experience, the system will undergo iterative development according to the Build-Measure-Learn framework,[26] starting with a small cohort (N=25) for preliminary usability and adoption testing, followed by further refinement and then testing with a larger cohort (N=200) to assess effects on the caregiver experience and in caregivers of diverse backgrounds. The end-product will be a high-impact AI care planner for ADRD that is service-ready, commercially viable, and capable of quickly scaling nationwide.

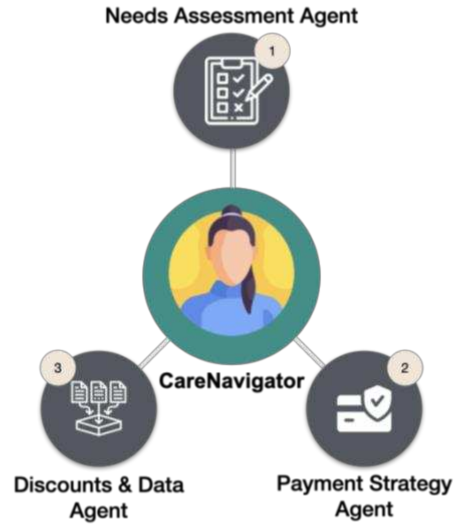

Figure 1: Illustration of the Olera CareNavigator and the 3-agent model

Value of SBIR Project. The value of the project lies in its potential to impact caregiver burden head-on, directly reducing its prevalence among the 61 million with disabilities and their 54 million family caregivers (11 million AD/ADRD caregivers in the US).[27] This project is the natural continuation of our preliminary work that identified the most prominent and critical challenges in AD/ADRD caregiving and the features caregivers most value in dementia care planning technology.[10,28] Our caregiver-centric Build-Measure-Learn approach significantly de-risks the proposed technology by involving caregivers directly in the design and confirming the "product-market" fit (i.e., high demand) early on in development. Our vision is to create a more resource-connected aging population in the US, and the economic value, if successful, is significant. For example, the technology holds the potential to help older Americans reclaim a significant portion of the $30 billion in under-utilized aid funding for the elderly while potentially saving the country billions in reduced utilization of healthcare services due to a more resource-connected population with basic needs met more consistently. Further, the project addresses multiple research gaps identified during the 2023 National Research Summit on Dementia Care, Services, and Support for People Living With Dementia and Their Caregivers/Care Partners.[29] Specifically, the development of technology that predicts disease trajectory and aids caregiver decision-making (G&O 7.2), exploring health tech's impact on accessibility, quality, and costs for individuals with AD/ADRD and their family caregivers (G&O 4.1), and enhancing diverse data repository of care support resources covering formal and informal care settings (G&O 4.1).

B) Expected Outcomes of the SBIR Project and Impact

Expected Outcomes of SBIR Project. The end product will be a high-impact AI care planner that has advanced functionalities to help 1) find affordable care solutions to needs and 2) better utilize resources, services and aid programs. Expected outcomes include multiple novel end products:

- **Novel Specialized AI Agents for Elder Care Planning**. We will create a novel Multi-Agent Network of specialized eldercare planning agents leveraging LLMs and natural language processing (NLP) techniques to provide tailored guidance to caregivers focusing on increasing accessibility to affordable care and aid programs. Our AI technology will be designed to assess functional requirements (utilizing a biopsychosocial framework) to predict care requirements and estimate care service costs, products, and other long-term care support mechanisms like aid programs.
- **Engaging and Widely Accessible User Interface / User Experience (UIUX).** Users will engage with a user interface featuring an engaging and realistic conversational AI Avatar (Figure 2) and website dashboard for articulating their needs, asking questions, and receiving personalized guidance – resembling a conversation with professional social workers during an initial or follow-up encounter. The AI conducts needs assessments, offering step-by-step instructions for using resources, cost-saving programs, and caregiving services. Clear instructions, devoid of jargon and information overload, inform decision-making. Written records of encounters and recommendations are provided in the web app dashboard for further ease of use and to archive the patient-caregiver journey.

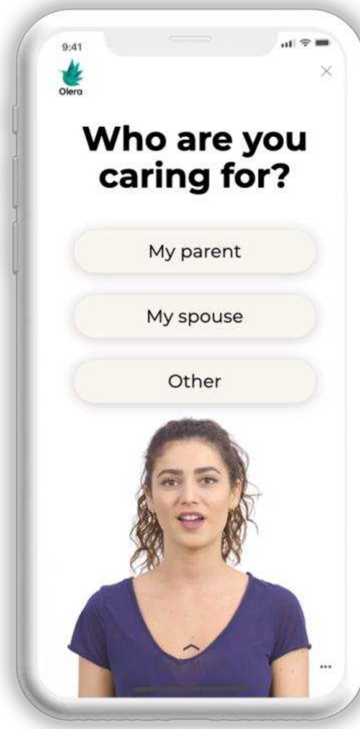

Figure 2: Representation of the Olera CareNavigator User Interface

- **Curated Educational Materials and Caregiver Resources (EMCR) Database.** Olera, Inc.'s EMCR database is central to the novelty and impact of the project – a continuation of prior SBIR work centralizing and standardizing data AD/ADRD caregiver resources. This expansive repository consolidates diverse care resources helpful in varying scenarios, including traditional (i.e., hospice or palliative services, home nursing, or wound care) and non-traditional care (i.e., non-medical home aid services, transportation, Elder Law Attorney, etc.), aid programs on the local, state, and federal levels (i.e., SSI, SSDI, SNAP, Medicaid, etc.), and even lesser known like support systems like community-based volunteer programs, adult day care programs, peer support groups among others.

Impact. The widespread adoption of this innovative technology could significantly reduce caregiver burden nationwide, benefiting over 11 million AD/ADRD caregivers and their care recipients. Broader impact potential stems from extending the reach to 61 million individuals with disabilities (and their 54 million family caregivers) who benefit from similar aid programs and affordable care solutions.

Financial Impact. Rising AD/ADRD caregiving expenses strain national resources. The Olera CareNavigator offers substantial relief, potentially yielding savings for individuals and society. For example, many older adults could save hundreds monthly by accessing aid programs like the Supplemental Nutrition Assistance Program (SNAP), a program averaging a $105 monthly benefit for food. Sadly, 5 million eligible older adults miss out due to enrollment confusion and not knowing where or how to apply, leaving $6.3 billion in unclaimed food assistance.[2] Those elderly adults eligible for Supplemental Security Income (SSI) could receive a monthly benefit of $914 as an individual or $1,371 for couples, but the application process takes 3-5 months, and many are unaware of the program or how to apply.[30] Mass eligibility matching with the Olera CareNavigator could quickly solve this and help Americans access a significant portion of the **$30 billion in unclaimed aid spanning essentials like food, prescriptions, utilities, and healthcare**.[2] There are similar opportunities like this that Olera, Inc. is exploring to match families with, including Medicare Part D Low-Income Subsidy for prescription savings ($10.6 billion yearly in unclaimed aid) and Medicare Savings Programs for copays ($4 billion yearly in unclaimed aid).[2] This technology could better connect AD/ADRD communities to critical aid programs, relieving financial strain and helping afford essential care to meet basic needs. Fostering a more resource-connected AD/ADRD community will improve overall well-being and health outcomes while helping to alleviate a portion of the annual $344 billion expenditure on AD/ADRD[5] care spent on costly emergency room visits and hospitalizations. This is a big problem in many hospitals across the country, and better resource utilization at scale could significantly reduce unnecessary hospital stays that could be "avoidable," which are typically not reimbursable by insurers. Hence, hospitals are at a loss financially, leading to downstream consequences on quality of care provided.

Broader Impact. The impact of this technology extends not only to the AD/ADRD community but also to the 61 million individuals caring for a family member with disabilities that require increased caregiver burdens. Like AD/ADRD caregivers, these caregivers would benefit from facilitated connections to resources and in finding affordable care options (the cost-saving of the CareNavigator are shown in Figure 3). Further, the CareNavigator, at its fullest potential, could bring information to people who have never been able to access higher levels of care planning due to barriers like transportation or accessibility. This reach will be particularly impactful in underserved communities where access to information and support for caregiving is more limited, however smartphone and internet accessibility remain prevalent as a delivery method for support.

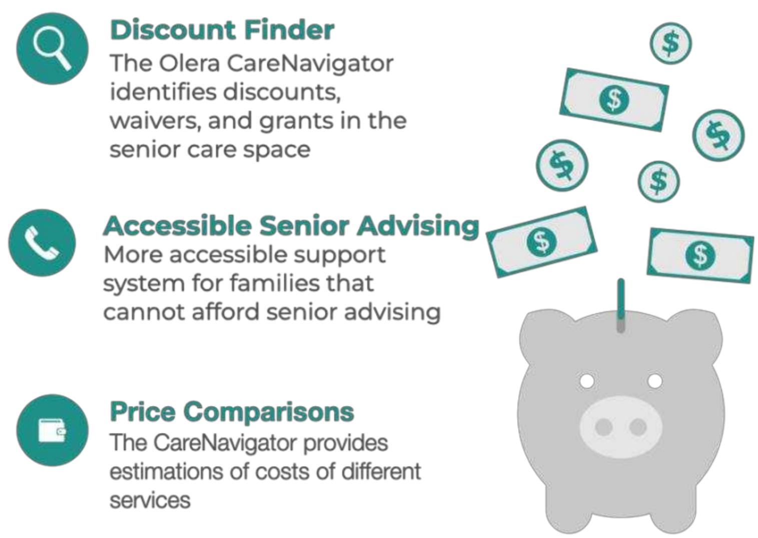

Figure 3: Cost-saving technology features of the Olera CareNavigator

Commercial Impact. At Phase IIB's conclusion, the Olera CareNavigator will be commercially viable and poised to scale to meet the demand for personalized care planning tools. It caters to a vast caregiver population, including those tending to AD/ADRD or other disabilities (~61 million total), thus addressing a notable market need within a substantial total addressable market (TAM). Our revenue potential is significant, establishing ourselves as a valuable “highway” for caregivers connecting to providers and products in the elderly and disabled services industry, currently valued at $64.4 billion.[31] The Olera CareNavigator leverages the existing Olera Marketplace, a digital platform from previous SBIR work that connects caregivers with care resources, providers, and products. It acts as a catalyst, redirecting caregivers to this marketplace when appropriate and fueling revenue streams (e.g., transaction fees, advertising, and memberships). **Importantly, our technology remains free for family caregivers, with revenue instead generated from corporate partnerships and affiliates.**

C) Market, Customer, and Competition

Market. The Olera Marketplace is two-sided, consisting of family caregivers seeking senior care solutions and businesses (or organizations) that offer products, services, and resources (illustrated in Figure 4). With an estimated 61 million American adults caring for family members with disabilities (11 million caregivers providing care to individuals with AD/ADRD) and the elderly & disabled services industry valued at $64.4 billion, these two expanding market segments represent significant growth opportunities for innovative solutions like ours. With AD/ADRD cases alone set to triple by 2050 worldwide, the demand for care services and products will only increase. To tackle the upcoming surge in demand, innovative and cost-effective solutions like the Olera CareNavigator become crucial for maintaining a sustainable support infrastructure for adults with disabilities and their caregivers.

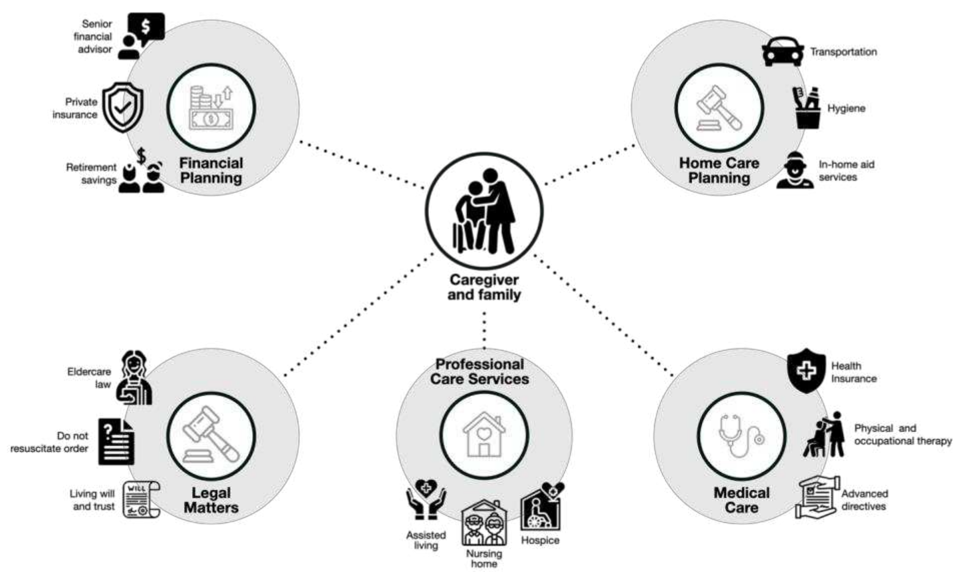

Figure 4: Services in the senior care ecosystem

Traction. In just the first few months since launching our Olera Marketplace in the Fall of 2022, we received an enthusiastic reception from family caregivers validating prior Fast-Track SBIR development efforts and serving as preliminary work showcasing the merit for this Phase IIB project. Below is a summary of milestones that we have met showing strong demand for care planning technology:

- Over 250 caregivers (mainly in Texas) have signed up for our current platform, recruited from a combination of word of mouth, digital marketing, and Search Engine Optimization (SEO), with our app earning a 4.57/5 average rating on the peer-reviewed Mobile Application Rating Scale in Phase I usability testing.[28]
- Our website traffic to the marketplace has climbed 30% quarter-over-quarter, from 1,300 new users in April 2023 to 1,700 by August, with organic traffic increasing by roughly 100% over that period. We expect to see strong growth as we become more established in our customer acquisition channels.
- We have rapidly onboarded over 30,000 service providers, building a national network hosted on our website with the proprietary Educational Materials and Caregiver Resources (EMCR) repository.[32]
- Over 10,000 family caregivers have joined the Olera Healthy Aging Community page on Facebook.

This strong user adoption and engagement across caregivers and industry providers demonstrates the high demand for dementia care resources and community support. As we add greater personalization with care planning functionality and new cost-saving features to the Olera CareNavigator using generative AI techniques and Large Language Models in Phase IIB, we aim to sustain this momentum, accelerate the growth of these successful traction activities and empower many more caregivers and corporate partners nationwide.

Customer. After over 200 interviews with family caregivers and industry service providers during prior SBIR work, we've identified two primary customer segments for the Olera Marketplace: 1) family caregivers and 2) Professional Care Assistance Organizations (PCAOs).

Family Caregivers. The first customer segment consists of caregivers of older adults needing long-term care, particularly those with AD/ADRD. Over the past two years, we have developed a community of over 6,000 members that fall into this customer segment, and over 250 family caregivers in Houston, Texas, are using our current web-based care planner (The Olera Platform). Caregivers come from diverse age groups and cultural backgrounds, with 6.1 million caregivers aged between 18-49 and 41.8 million aged 50 and older.[33] Throughout the previous Fast-Track SBIR, we conducted extensive usability assessments to ensure the current user interface that hosts the Olera Marketplace is adoptable across various age groups and backgrounds. The interface was tested using the Mobile Application Rating Scale (MARS), a popular measurement tool used to evaluate functionality, design, information quality, and engagement of health-related smartphone applications on a 5-point scale survey. The caregiver support platform scored 4.6 among caregivers of all ages (with a minimum goal of 3.6). We attribute this success to the well-established Build-Measure-Learn framework,[26] and as we move into Phase IIB, the team remains committed to maintaining the same development and usability standards. Our current marketplace platform will integrate with the new Olera CareNavigator in an inclusive and accessible manner for caregivers from diverse backgrounds, considering variations such as ethnicity, race, language, age, literacy level, and technical abilities.

The second customer segment comprises Professional Care Assistance Organizations (PCAOs), encompassing various businesses and professional entities that support older adults and caregivers. Some of these services include home care agencies, long-term facilities, financial planners, and elder law attorneys. After over 200 interviews with these organizations under the mentorship of NSF and NIH Innovation Corps programs, we found they are in competitive markets with large amounts of their operating budget allocated to customer acquisition. The competitive nature of the industry incentivizes PCAOs to raise awareness of their services and positions our audiences as a highly valuable opportunity for advertising. Our technology also allows new and emerging local businesses to advertise their services to expedite their entry into the market and meet the growing needs of eldercare. This is particularly true for newly established senior care services that need deeper networks and brands to attract new customers.

Existing Solutions & Competitive Differentiation. When tackling dementia care challenges, caregivers commonly rely on healthcare professionals, family, friends, and the internet.[10] Despite their significance in the caregiving journey, these resources don't always provide practical assistance for caregivers' primary challenges—identifying, affording, and utilizing essential care. For instance, due to time constraints, healthcare providers' assistance is often limited to disease-related information and treatment insights.[34] While friends, family, and support groups offer alternative information and social support sources, their availability is inconsistent, and the information can be subjective and incomplete.[35] Expert websites like NIH, CDC, Alzheimer's Association, and National Council on Aging provide vital information for AD/ADRD caregivers. However, these static web pages lack personalization and are not region-specific, making it time-consuming and frustrating to sift through irrelevant information.[36,37] The problem is exacerbated because most caregiving information online inundates readers with a plethora of information that is overwhelming and hard to translate into practical steps to take. This challenge is heightened for individuals with barriers such as low literacy and technical skills.[18] Ultimately, there are no widely adopted, comprehensive tools for personalized care planning or resource connection for families seeking guidance on affording and initiating necessary care.[10] Compared to numerous online senior care referral services like A Place For Mom, Caring.com, and Seniorly.com exist and in-person geriatric social work or free placement agencies. Olera differentiates its value to caregivers through several key factors (and others shown in Table 1):

- **Accessibility and Affordability.** Olera focuses on enhancing affordability and accessibility in senior care services. It identifies discounts, aid and benefits similar to senior advising interventions but at no cost and on-demand, and empowers users with transparent pricing information and financial strategies to address expenses.

- **Comprehensive Guidance.** Olera extends beyond senior living, providing personalized care planning and recommendations, leveraging its robust EMCR database to connect users with a wide range of services.
- **AI-Personalization.** Olera employs AI for advanced personalization, tailoring recommendations to caregivers' and care recipients' needs and financial situations.
- **Privacy Protection.** Olera prioritizes user privacy, allowing caregivers to control the sharing of personal information with service providers (see privacy section) and preventing unsolicited calls and service marketing.
- **Accessible User Interface.** Olera's user interface is rigorously tested and designed for caregiver efficacy, reducing burden and improving decision-making confidence.[28]

	Olera Care Navigator	Geriatric Care Managers	aPlaceforMom	Alzheimer's navigator alzheimer's association
Personalized Care Guidance	✓	✓	✗	✗
Cost Savings Finder	✓	✓	✗	✗
AI-Personalization	✓	✗	✗	✗
Privacy Protection	✓	✓	✗	✓
Free for end-users	✓	✓	✓	✓
On-demand support	✓	✗	✓	✓

Table 1: Existing Solutions & Competitive Landscape

D) Revenue Streams

<u>Revenue Streams.</u> The Olera CareNavigator fuels existing revenue channels in the Olera Marketplace, which employs two primary monetization strategies: 1) corporate partnerships (and lead sales) and 2) affiliate marketing fees. The quick scaling of the CareNavigator post-Phase IIB will drive users to the Olera Marketplace, where they can connect them with corporate partners and affiliates as needs arise. This strategy drives significant revenue generation opportunities and fuels growth for Olera, Inc. because of the valuable audience it creates within the industry. As Olera CareNavigator establishes itself, diversification of senior care offerings through affiliate marketing, strategic collaborations with industry providers and stakeholders, and data-driven insights for partners are avenues for expansion. Further validation and iteration on this revenue model will be developed in collaboration with industry advisors such as David Chandler (letter of support attached), Senior Director of Strategic Programs at Senior Helpers – a nationwide home care provider, who will work with us to ensure our revenue strategy brings value to the senior care industry.

<u>Corporate Partnerships</u>. PCAOs eligible for corporate partnerships can be categorized into home care agencies, senior care facilities, and non-profit PCAOs. The distinct nature of offerings in each sector yields varied economics, with some, like in-home care services, having high operating costs, slim profit margins, and limited customer acquisition budgets. Conversations with in-home care providers reveal a willingness to invest $30-$50 in client acquisition, highlighting the demand for cost-effective marketing solutions validated through customer discovery interviews. Conversely, more established senior living providers with high-profit margins allocate

significant budgets for customer acquisition due to the high lifetime value of a customer. For example, a private room in a nursing home costs around $114,007 annually, prompting senior living providers to invest $3,000 - $5,000 on average to acquire a single customer sale. Because of this, we can charge a rate to providers for “leads” generated on our site that may convert to business for them. Figure 5 shows the lead generation process.

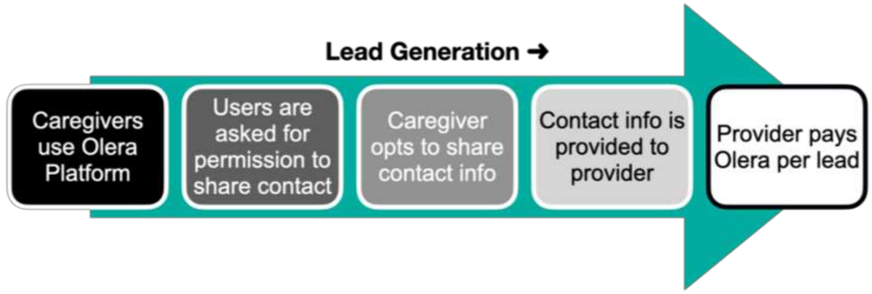

Figure 5: The lead generation process

Affiliate Product Marketing. Olera can generate revenue through affiliate marketing of in-demand senior care products, earning commissions on sales when caregivers purchase from our site. For example, fall-alert bracelets offer a 15% commission per $100 sale, while home safety products yield a 10% commission on average for a $200 order. Senior-friendly communication devices provide up to 8% commission for a $150 product. With a large user base and growing search presence, this affiliate marketing can yield substantial revenue over the long term.

Revenue Projections. Moreover, we expect a substantial increase in projected revenue from the previous commercialization plan in past Fast-Track SBIR work due to the increased synergy the Olera CareNavigator introduces. This is due to the superior resource-matching capacity that AI introduces, making leads generated by the marketplace more likely to convert to sales for affiliates due to better pre-qualification upfront. Because of this, Olera estimates that industry providers will find our leads more valuable, and in turn our lead sale conversion rates could increase anywhere from 8-32% we predict. For modeling, we’ll use an estimated increase in our lead conversion rate by 17.5% from 25% in our previous SBIR proposal to 42.5%. **This increase in conversion rates would result in a projected revenue of $1,041,675 in 2025, increased from ~$600,000 previously projected without the Olera CareNavigator (Table 2).**

Earrings Projections are based on the following assumptions:

- ~80,000 end-user caregivers visiting the Olera Marketplace in the year 2025 (0.5% of 16.34 AD/ADRD caregivers in the US)
- 2% annual end-user base growth rate
- Three leads (monetizable units) generated per end-user per year
- Lead sell rate at 42.5% (increased from 25% due to the Olera CareNavigator development and Launch)
- Price per lead is an average of $10 per lead

		Expected TAM Captured (2% annual growth rate)			Estimated Caregivers Using Olera		
	Number of Caregivers	2025	2026	2027	2025	2026	2027
AD/ADRD Caregivers	16,340,000	0.50%	2.50%	4.50%	81,700	408,500	735,300
Leads Generated (3 per caregiver)					245,100	1,225,500	3,676,500
Leads Sold **(42.5% conversion rate)**					104,168	520,838	1,562,513
Projected Earnings (conservative $10 per lead sold to industry provider)					$1,041,675.00	$5,208,375.00	$15,625,125.00

Table 2: Revenue Projections

E) Production and Marketing Plan

Production Plan. Our production plan centers around three specific aims vital to the successful development of the Olera CareNavigator (Figure 6). The first aim is dedicated to the development of each specialized agent.

This 12-month effort will be led by Dr. Jim Nolan, Ph.D. (biosketch and letter provided), an expert on developing Large Language Models (LLMs), deep learning, word embeddings, and transformers. Jim has previously worked with the National Institute of Allergy and Infectious Diseases (NIAID) and the National Institutes of Health (NIH) on developing LLMs to detect misinformation about vaccines and infectious diseases showing his commitment to applying AI to healthcare. In Aim 2, these agents will be integrated into a Multi-Agent Network and trained using insights provided by domain experts in social work (licensed clinical social workers support letters attached).This Network will then be integrated into an intelligent user interface with custom life-like 3D avatars that simulate human interaction using generative AI frameworks from Synthesia.io and Runway MI. Insights gathered from preliminary user testing of the UI will guide further refinements, ensuring universal effectiveness, adoptability, and cultural relevance in future product cycles aligned with the Build-Measure-Learn framework.[26]

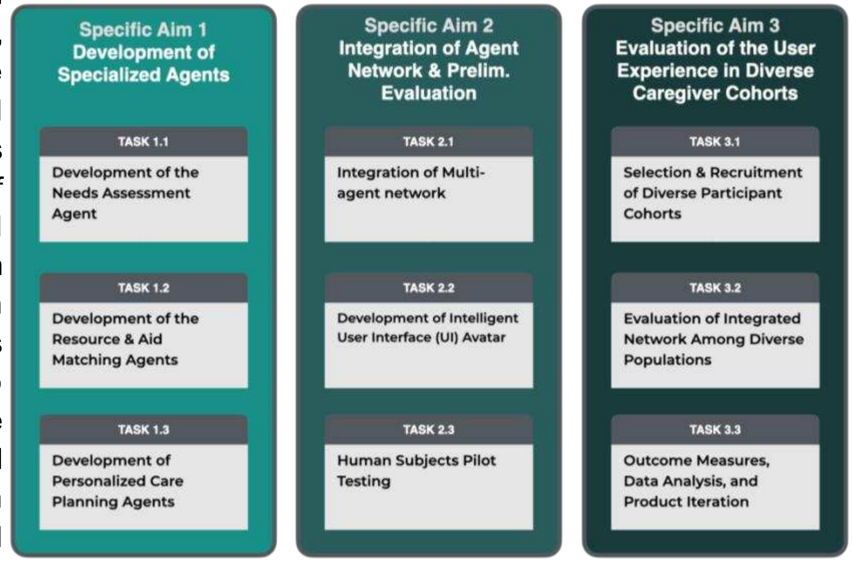

Figure 6: Overview of Project Plan for Phase IIB

Data Privacy, Safety & HIPPA Compliance. Olera is committed to maintaining the highest data privacy, safety, and HIPAA compliance standards with its innovative offerings. The platform will collect personal or sensitive user data due to the nature of elder care planning. During development, user data collected by the technology will be handled according to our Data Safety Monitoring Plan as outlined in our study design and all activities with human subjects will be approved by the Clemson University IRB. We will obtain professional legal counsel regarding handling HIPPA-sensitive data, and best-practice precautions will be put in place before commercial launch. We will also explore legal counsel to put the proper mechanisms in place for “Release of Information” protocols to safely streamline resource referral logistics. Finally, by using Reinforcement Learning from Human Feedback (RLHF) and collaborating with experts, we plan to ensure and accuracy of our technology thorough in-house testing and third-party audits before public release.

Marketing Plan. Our marketing efforts are thoughtfully tailored to our two primary end-user groups: (1) family caregivers and (2) professional care assistance organizations (PCAOs). The marketing channels and strategies to reach each group are detailed below.

Family caregivers. Over three years, we've explored diverse marketing channels and tested campaigns, achieving notable success. The most effective and cost-efficient approaches led to a steady monthly growth rate of ~10%. This growth is set to accelerate with ongoing iterative product launches and continued focus on the following:

- **Community Building.** Though senior care and aging are rarely highlighted in our society, millions of older adults and caregivers confronted with these topics are seeking a compassionate community. We established the "Healthy Aging Community," an online space honoring seniors and caregivers to bridge this gap. This community has flourished, boasting 10,000+ registered members, and is our most significant marketing channel. Its expansion, fueled by events, social media, and education, creates a network effect as members engage, form bonds, and invite new members.

- **Organic Traffic Generation Through Content Marketing, Digital PR & SEO.** The nature of our platform involves the generation of a large volume of content housed in our Educational Material and Caregiver Resource (EMCR) repository. The web app developed in Phase I and II currently features over 6,000 indexed web pages, including educational articles, local senior care resources for cities nationwide, and listings of providers from various industries. Although initially difficult to populate and organize, once created, this trove of content serves as a driver of organic traffic to our platform from users searching for relevant information and resources. We have seen promising results from this strategy and currently receive roughly 400 monthly visitors organically without any advertisement spending. This number will only increase as quality educational content is added to the EMCR, a strategy heavily invested in by the company. To further increase growth through this channel, we will identify the areas of greatest need for caregiver education and develop a robust content strategy on these topics that is helpful to users while being optimal for driving organic traffic to our platform. The content is amplified via digital PR outreach to relevant journalists, podcast hosts, and top bloggers. This connects Olera with a broader audience and stimulates trust and authority within our sector. This strategy drives our advertisement costs down significantly over time and grows a healthy, reliable community. Our SEO strategy is aided by seasoned consultant Simon Cowan (letter of support attached), who has experience scaling brands and products and is committed to leading the team to build traction for the Olera Platform and CareNavigator.

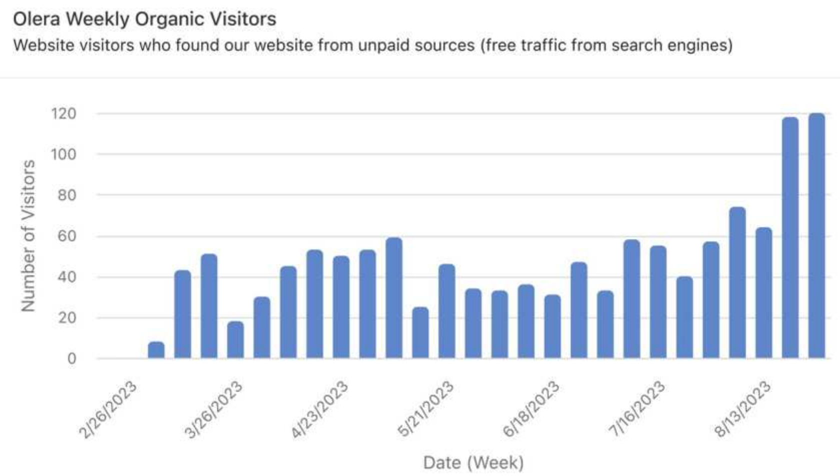

Figure 7: Growth in weekly Organic Traffic to the Olera website (Feb - Sep 2023)

- **Promotion from Key Partners.** We plan to continue cultivating relationships with strategic partners with extensive caregiver networks to connect to more caregivers. Existing collaborations with the Alzheimer's Association, Texas A&M Center for Population Health and Aging, and Clemson University have effectively showcased our platform to their audiences. We aim to expand this network, encompassing local Area Agencies on Aging, senior centers, caregiver support groups, and industry entities (home care, hospice or palliative agencies, senior living facilities, elder law attorneys, and geriatric care managers). These partnerships hold significant potential for user influx, with approximately 5,000 new AD/ADRD caregivers expected within the first year. Additionally, we're targeting national entities like AARP and the Aging Life Care Association, both strong brands aligning with our mission.

PCAOs and Corporate Partners. Given the competitive nature of the senior care industry, service providers and product owners are incentivized to seek channels to reach more family caregivers to expand their marketing reach. Additionally, most organizations publicize their contact information on their marketing websites, making networking straightforward. Therefore, marketing to this market segment is much easier than family caregivers. To expand our reach of service providers, we run promotions for new providers, such as free advertising and spotlight features for providers that meet a high standard of service. We also plan to attend industry conferences and networking events to build relationships and expand our network of providers, especially during the early product life cycle of the Olera CareNavigator. By running targeting email marketing campaigns that outline our platform's unique values, we have seen a steadily increasing number of providers in our network. Early PCAO marketing efforts have demonstrated that this is a viable growth strategy.

G) Company Overview

Olera, Inc. is on a mission to address one of the largest problems facing our society: the inability of a large percentage of Americans to find and afford senior care support. We aim to leverage novel software technologies to alleviate the growing caregiver burden and enhance the well-being of older Americans. Our core competencies are developing user-friendly and caregiver-focused platforms with AI, database management, and software interfaces optimized for diverse populations.

Leadership Team. The company is led by CEO **Tokunbo "TJ" Falohun, MS,** a trained biomedical engineer and serial medical technology entrepreneur. As a former engineer at Pfizer and a recipient of the National Science Foundation Graduate Research Fellowship, TJ is experienced in the technical development of novel digital technologies to improve human health. Having co-founded multiple med tech companies that have received SBIR awards and outside capital, TJ has developed critical entrepreneurial leadership skills and was inducted into the Texas Business Hall of Fame as a future innovative business leader in Texas. He has since specialized in AI for digital health applications and holds a patent for detecting ocular diseases using deep learning. Now dedicated to the company full-time, TJ supervises teams, aligns tech development, and leads outreach.

Co-founder **Logan DuBose, MD, MBA,** the COO of Olera, Inc., contributes real-world experience in clinical care for AD/ADRD patients and working with family caregivers, social support services, and healthcare administration. Dr. DuBose also manages human subjects research, including research design, recruitment, data collection, and operations. Leveraging business acumen for the venture, he also works closely with TJ on outreach and overall vision for the company and its products. Critical to the founding team, Interim CTO **Jesse Phipps, MS,** skilled in computer engineering and artificial intelligence, oversees technology development, and Interim CMO **Jeswin Vennatt, MBA** and MD Candidate, spearheads start-up traction efforts. The team is assisted by accomplished R&D Engineer **Diana Salha,** who aids funding acquisition and vision planning with expertise in public health.

This dynamic leadership team is complemented by a rich network of external collaborators – experts in the senior care industry, aging research, Large Language Model (LLM) development, private equity, marketing, and entrepreneurship. Their collective wisdom propels us forward. **Dr. Marcia Ory, PhD, MPH,** brings over two decades of experience in aging research. **Dr. Jim Nolan, PhD.**, renowned for language model development and has previously worked with SBIRs from the National Institute of Allergy and Infectious Diseases (NIAID), adds to our AI and NLP capabilities. **David Chandler** offers industry guidance, and Dr. Qiping Fan and **Dr. Geong Han** provide invaluable methodology advice with biostatistical design and analysis. A seasoned digital marketing veteran, **Simon Cowan** shapes our marketing strategy as lead advisor for content and affiliate marketing, digital PR, and search engine optimization.

Future Direction. As we move forward, we remain steadfast in our commitment to leverage novel technologies to improve the affordability and accessibility of care for caregivers and older adults. Our trajectory is set for growth with our current Olera Platform web app rated 4.57/5 on the Mobile Application Rating Scale and a rapidly growing user base. Our proprietary Educational Materials and Caregiver Resources (EMCR) repository now hosts a robust network of over 5,000 service providers (expected to be 15,000 by Spring 2024). Simultaneously, our Olera Healthy Aging Community has drawn over 10,000 family caregivers (expected to be 20,000 by Spring 2024) seeking support and information. With a solid foundation of achievements and an unwavering leadership team, we are poised to further develop our care planning technology, amplify our impact, and continue our journey toward shaping a more positive and connected aging experience for all.

Finance Plan. Olera, Inc. has been financed by non-dilutive funding through Texas A&M, the National Science Foundation, Blackstone Techstars, and the National Institute on Aging SBIR program, totaling $2.7 million. Continued support from the SBIR program in Phase IIB will help fund critical technology development efforts, allowing families to identify relevant financial support programs. During these 24 months, we will also be laser-focused on growing traction among caregivers and PCAOs according to the plans described in the marketing section. After demonstrating steady user growth and refining our revenue model, Olera plans to raise $10M in a Series A round at a $50M post-money valuation to further Olera's mission. **So far, we have received strong interest from investors, such as the Aggies Investor Network (letter attached), to start a Series A round if milestones for Phase IIB are met. We also received similar interest from private equity firm leader Jason Halsey (letter attached).** Revenue generation will supplement further growth once commercialization is achieved, which we expect to be on schedule or earlier than expected during Phase IIB development.

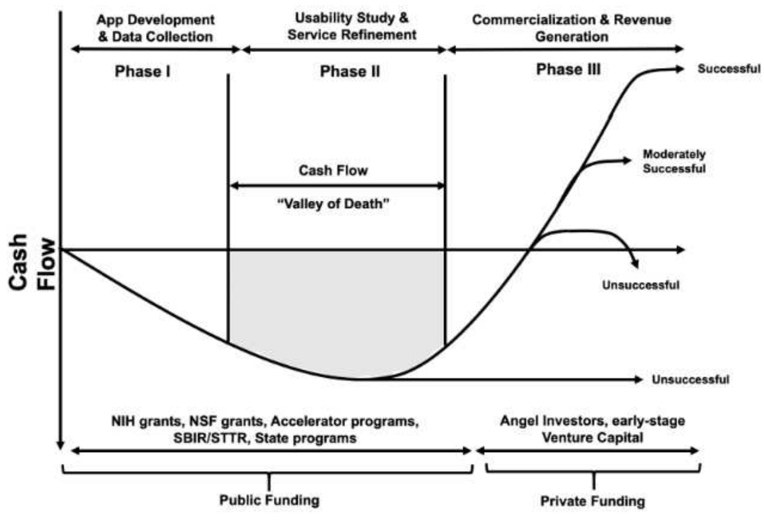

Figure 8: Olera Financing Plan

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Progress Report Publication List

To date, we have had the opportunity to disseminate findings as below and plan to continue to publish results as data becomes available:

Development of a personalized digital support platform for AD/ADRD caregivers from diverse populations: Caregivers of people living with Alzheimer's disease (AD) and AD-related dementias (ADRD) face a high degree of psychological, physical and financial stress. Such caregivers face a higher risk of developing depression, anxiety, social isolation, and physical health problems due to the chronic stress associated with caregiving. To address this challenge, we developed a digital platform that improves the access to tailored information and educational resources to assist with the financial, legal, psychological, and estate planning challenges associated with dementia care. I served as the co-Investigator for this study.

DuBose, L., Falohun, T., Fan, Q., Hoang, M-N., Doyle, D., Kew, N., Lee, S., Ory, M. (Year). Development of a tailored digital platform to assist caregivers of persons living with dementia: Usability study. Presented at the Public Health Center for Excellence in Dementia Caregiving BOLD Conference.

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SCHOOL OF PUBLIC HEALTH
HEALTH SCIENCE CENTER

Department of Environmental and Occupational Health

TEXAS A&M UNIVERSITY

September 1, 2023

Dear Principals of Olera Inc.,

I am pleased to offer my support for your SBIR application and see this as an opportunity to continue building upon our previous collaborative efforts in delivering personalized resources and support technology for family caregivers. The preliminary work we have accomplished together has underscored the urgent need for a digital tool capable of assisting caregivers, and your proposed Olera CareNavigator appears well-positioned to effectively address the specific challenges we have identified. The encouraging outcomes from our past SBIR endeavor, which have already demonstrated the potential of delivering tailored caregiving resources, are a testament to the promise of your approach. I am particularly impressed by your engagement of such a large community of family caregivers already during the developmental phase of the existing Olera Platform.

Moreover, your strategic approach to commercialization outlined in your proposal further highlights your readiness for subsequent stages and ongoing advancements in your care planning technology. I am eager to see how integrating artificial intelligence enhances resource matching and delivers impactful information that can positively influence caregiver behaviors. Ultimately, I am steadfast in my commitment with you in creating the Olera CareNavigator to enable a more well-equipped caregiver population.

As my biosketch indicates I have extensive research and practice expertise in aging and health care, especially around dementia caregiving. Building on more than a 40-year career as a leading social gerontologist, I will commit some of my dedicated consultant time to work on this project which will be mutually beneficial to both academia and industry. My proposed role as the lead academic advisor entails high-level guidance on the human subjects research component of your project. This will involve providing overall scientific direction, working closely with Dr. Qiping Fan at Clemson University, a previous student of mine and current collaborator on our previous SBIR and leveraging the expertise of Dr. Han, a long-time colleague from the School of Public Health at Texas A&M University and an expert in biostatistics. Specifically, I will offer my expertise to ensure the implementation of high-quality human subjects research that rigorously tests the effectiveness of your technology. This will involve advising on research methodologies, subject recruitment, data collection, analysis, and interpretation of results. I am confident that my involvement will contribute to the robustness of the research and the overall success of the project.

Thank you for the opportunity to be a part of this important project. I am eager to lend my support and expertise to guide the development of the Olera CareNavigator, which has the potential to make a meaningful difference in the lives of caregivers and people living with dementia.

Sincerely,

Marcia G. Ory

Marcia G. Ory, PhD, MPH
Regents & Distinguished Professor

360 SPH Administration Building
1266 TAMU
College Station, TX 77843-1266

Tel. 979.436.9368
mory@tamu.edu

MAYS BUSINESS SCHOOL
McFerrin Center for Entrepreneurship

Blake Petty '98
Executive Director | McFerrin Center for Entrepreneurship
Executive Director | Aggie Angel Network
aggieangelnetwork.com

TEXAS A&M UNIVERSITY

August 30, 2023

To: Review Committee Members et al

RE: Letter of Support for Olera, Inc.'s SBIR proposal in collaboration with Texas A&M University and Clemson University

I am delighted to provide this endorsement letter for Olera, Inc.'s referenced SBIR proposal. As Executive Director of the McFerrin Center for Entrepreneurship in Texas A&M University's Mays Business School, I remain particularly excited about this project's commercialization potential.

The McFerrin Center's primary goal is to facilitate the growth and success of startup companies. The Center utilizes over 35 annual programs to help entrepreneurs explore, establish and expand their businesses, and provides them with connections to resources, advisors and a community to support their startups. The Center helps annually host the *Texas A&M New Ventures Competition*, which invites companies like Olera, Inc., to pitch their SBIR-stage opportunities for non-dilutive award funds (https://texasnvc.org/). Introducing Olera to the *New Ventures Competition* is just one example of the power behind connecting across Texas A&M's alumni base (the "Aggie Network") to bring together people with a common goal to support successful startups.

Another program in the McFerrin Center is the *Aggie Angel Network* (AAN), for which I serve as founder and Executive Director. Launched in 2010 as one of the first university-affiliated angel networks, AAN has successfully placed over $16MM into 53 startup companies seeking seed-stage investments (http://aggieangelnetwork.com). I absolutely consider Olera, Inc., a prime candidate for invitation to pitch to AAN in an upcoming investment cycle, following completion of their SBIR Phase 2b milestones.

Their team's vision toward developing specialized AI tools for Elder Care planning certainly resonates strongly within our local investment community, and their strategic utilization of the nationwide database of senior care resources they have built and continue to expand during Phases I and II offer tremendous potential for lead-referrals and an attractive business model in a large and growing market.

I wholeheartedly support this application for funding and will continue to introduce the company to others across the Aggie Network who can continue to help advise and fund the company as it matures.

Sincerely,

Blake Petty

4123 TAMU | College Station, TX 77843-4123
Physical: 1700 Research Pkwy, Ste 120 | College Station, TX 77845
Texas A&M University Research Park
blakepetty@tamu.edu

ENTREPRENEURS MADE HERE entrepreneur.tamu.edu

To the Principals of Olera Inc.,

I am writing to express my strong support for this Phase IIB application to develop the Olera CareNavigator and to reiterate my commitment to my role as the Lead Technical Advisor for this innovative project. The potential impact of the Olera CareNavigator on senior care, caregivers, and individuals affected by Alzheimer's disease and related dementias (AD/ADRD) is something I am committed to solving. I am excited to leverage my extensive experience in Natural Language Processing (NLP), Machine Learning (ML), and Artificial Intelligence (AI) and contribute to solving one of our country's most pressing challenges – senior care costs and access.

Our preliminary work together has already been instrumental in demonstrating the feasibility and technical soundness of the Olera CareNavigator approach section detailed in the research plan. During the development phase described, we will further iterate on the prototype and integrate the EMCR training database but now optimizing it with family caregiver and domain expert input, building upon the initial eldercare-specific Large Language Model (LLM) capability, and showcasing the power of AI-driven solutions in the context of senior care and improvements in caregiver outcomes for the better. My role in this preliminary work, alongside my experience in machine learning technologies, has been crucial in establishing the foundation for the project's success both now and in the future.

I am most excited to continue working directly with founders TJ Falohun and Jesse Phillips, both accomplished early-career investigators with engineering accomplishments that have already established them as substantial innovators in AI. As the Technical Advisor, I am fully committed to building upon this strong team foundation to ensure the project's success. My experience in AI, including my work as Principal Investigator (PI) on NIH, DARPA, and other government laboratory efforts, positions me well to guide and assist directly with the development of the specialized AI agents and bridge the gap between academia and industry. This collaborative approach, with engineers TJ and Jesse, as well as the entire Olera Inc. team, will not only further validate our technical approach but also drive innovation and optimization.

In summary, our preliminary work, which demonstrated feasibility and addressed technical limitations, combined with technical expertise and collaboration with TJ Falohun and Jesse Phillips, and the interdisciplinary Olera team significantly de-risks the technical success of the Olera CareNavigator and I look forward to the exciting journey ahead with the Olera Inc. team.

Sincerely,

James J. Nolan, Ph.D.
9/2/2023

September 4, 2023

Dear Reviewers,

I am writing to endorse Olera's grant proposal to create the Olera CareNavigator and to emphasize my commitment to continue advising the team towards achieving their mission to connect caregivers with resources. I believe the Olera CareNavigator holds the potential to be a transformative force within the senior care industry. Olera's grant proposal represents a significant stride towards addressing the critical issues of access and affordability in eldercare, and I am eager to lend my extensive experience as a seasoned provider in the home care sector to contribute meaningfully to this endeavor.

During my time as Senior Director of Strategic Programs at Senior Helpers and my previous experience as an Executive Director for Retirement Center Management and Director of Clinical Services for Brookdale Senior Living, I have gained invaluable insights into the home care industry. I am honored to serve as an advisor to Olera's business development team, with a specific focus on the development of the corporate partner program and the exploration of revenue models. My educational qualifications, including a Master of Business Administration (MBA) in Healthcare Administration from Eastern University and a Bachelor of Science in Nursing (BSN) from Oral Roberts University, further bolster my preparedness to contribute significantly to Olera's initiatives from an interdisciplinary perspective that complements my experience working as a leader in the home care industry.

As Olera progresses through the grant proposal process, I am committed to actively collaborating with the entire Olera team. Together, we can refine the business strategy, fortify revenue-generation methods, and craft a compelling value proposition that benefits both family caregivers and senior care industry providers. My role as an advisor is centered around enhancing Olera's corporate partner program and exploring innovative revenue models to ensure the sustainability and growth of this mission-driven endeavor.

I would like to extend my gratitude for the opportunity to support Olera's grant proposal, and I eagerly anticipate our productive collaboration. Together, we can make a significant impact in addressing the challenges faced by the senior care industry.

Sincerely,

David Chandler RN, BSN, MBA

HALSEY | LEGAL
BUSINESS CONSULTING

September 1, 2023

Re: Olera Incorporated SBIR Proposal

To Whom It May Concern:

I am writing to provide my endorsement for Olera, Inc.'s SBIR proposal outlining their plans to develop an innovative artificial intelligence system to match families with essential resources for aging care. With a career spanning over two decades as business consultant, attorney, and currently as a leader within a private equity firm specializing in healthcare start-ups, I am excited about the potential of Olera's team and their proposal to introduce this high-impact and commercially viable technology.

I have been deeply engaged in advancing innovation and breakthrough solutions within the healthcare sector throughout my career. Olera, Inc.'s mission of enhancing the caregiving journey through innovative resource-matching aligns seamlessly with our vision as a private equity investing in high impact startups, and their proposed CareNavigator AI System embodies the kind of forward-thinking approach that resonates with our core value of funding disruptive technology in healthcare.

I firmly believe that Olera, Inc. stands as an exceptional candidate for inclusion in a forthcoming investment cycle upon the successful completion of their development of the CareNavigator and completion of milestones laid out in this proposal. The progress achieved through their SBIR work so far serves as a testament to their commitment and competence, further de-risking investment in this team. Founders TJ and Logan have a clear understanding of the problems at hand in senior caregiving having conducted thorough research into the matter and are forward-thinking in their innovative approach to technology solutions in the space.

Recognizing the pivotal role that advisory support plays in driving the development and commercialization of innovative technologies, I am eager to continue offering my expertise and insights to Olera, Inc. during the development of the CareNavigator and resources database (EMCR). My prior experience with SBIR initiatives equips me with a comprehensive understanding of the process and the nuances involved in steering projects towards their grant milestones, as well as positioning them for subsequent commercial viability.

Anticipating Olera, Inc.'s intention to incorporate private equity funding into their financing plans, I am eager to continue to advise and champion an invitation to pitch to our firm upon the successful culmination of Phase IIB. Their demonstrated potential to devise pioneering solutions to the pressing challenges in eldercare resonates strongly with our investment philosophy and I would like to continue to play a significant role in their success now and in their future.

Thank you for considering my endorsement. Should you require any further information or clarification, please do not hesitate to reach out to me.

Sincerely,

Jason Halsey

WWW.HALSEYLLC.COM (210) 728-6177

Dear Dr. DuBose,

I am genuinely excited to support Olera, Inc's proposal for specialized AI tools in Elder Care planning, aiming to mitigate costs for families and connect them to underutilized and critical resources at an important phase of life. This proposal has the potential to greatly benefit client systems such as families and patients dealing with dementia and other disabilities, populations I have worked closely with as a licensed social worker in the hospital, hospice, psychiatric hospital, high education environment, policy environment, and in academic pursuits as a PhD in Interdisciplinary Health Sciences at the University of Texas at El Paso.

As someone with front line, deep, and boots on the ground experience from healthcare and academic research about gerontology and the US-Mexico border, I see the urgency for innovative and effective solutions. The proposal's use of reinforcement learning from human feedback (RLHF) to train AI models offers an exciting opportunity to capture several computationally expensive and time consuming aspects of social work and case management that can be optimized and directed to technology such as resource identification, referral generation, eligibility matching, and more. Many social work functions can fall through the cracks because the client system does not have the time, patience, or expertise to navigate through complicated systems. This commercial technology and future innovations that come from it could potentially lead to more health equity and great population health outcomes due to reducing or eliminating barriers in completing healthcare behaviors.

Collaborating with Olera as a member of their panel of domain experts in social work aligns with my commitment to quality healthcare for all and disability justice. I also see a significant need for support for patients that speak Spanish in geriatric and dementia care so this commercial product could be easily translated into other languages and be trained by cultural consultants and client systems themselves. Drawing on my experiences since taking Nursing Care for the Aging Client in 2008 up until now, I intend to contribute value, consultation, and subject matter expertise during AI agent training on the RLHF panel outlined in the proposal.

This initiative project resonates with my professional social work values and mission to aid marginalized and high need populations overcome barriers to care such as financial and resource obstacles to quality care. I am confident that this project will improve caregivers' and families' lives across the nation and ultimately lead to better outcomes for people with disabilities.

Sincerely,

Nicolas D. Silva, LMSW
Ph.D. student in Interdisciplinary Health Sciences
Society for Neuroscience Neuroscience Scholars Program (NSP) Associate 2023-2025
Preparing Future Faculty Fellow 2023

Dear Olera Inc. Principals:

I am pleased to write a letter of support for your application on behalf of Clemson University Department of Public Health Sciences. Having worked with you since 2021, we are in full agreement that a digital tool to help caregivers of persons living with dementia better navigate the maze of information about financial, legal and other planning aspects of care is urgently needed. We are especially impressed that you have already engaged many stakeholders in your design development, and have laid the foundation for a practical and easy to use customized digital tool.

As your goals are clearly aligned the Clemson University Elevate Goals in doubling research and collaboration with industry, we have agreed to be serve as your academic partner. In particular, we have recently completed a study of caregivers and technology and can attest to the fact that dementia caregivers are eager to adopt technological solutions to ease their burdens of care.

I will serve as PI for the Clemson subcontract, and co-investigator for the overall proposed project. In addition to Clemson core resources, the Department of Public Health Sciences has local, state and national grant support from the Centers for Disease Control and Prevention, National Institutes for Health, the Agency for Health Care Research and Quality, and Prisma Health, to name a few. Our Department and College have multiple full-time staff to support research activities and more.

We see our involvement in several related core Olera, Inc. grant functions. We will contribute to overall research scope and framework for human subjects research, obtaining IRB approval from Clemson University IRB for all studies involved, consulting on the research design, including sample size, population targets, data collection strategies and measurement for human subjects research, conducting data analyses using tools, instruments, and scales to measure human subjects outcomes, assisting with report writing, publication, and research dissemination for all research and technology development.

The Department fully supports your application, and see us as a core team member leveraging our partnerships and existing resources to help amplify Olera, Inc. goals. We wish you success on this highly timely application that can bring financial and legal information to underserved dementia caregivers.

Sincerely,

Qiping Fan

Qiping Fan, DrPH, MS
Assistant Professor of Epidemiology
Department of Public Health Sciences
College of Behavioral, Social and Health Sciences
Office: 524 Edwards Hall, Clemson University, Clemson, SC 29634
Phone Number: 984-245-5244
Email: qipingf@clemson.edu

September 4, 2023

Dr. DuBose and Mr. Falohun,

In my capacity as a Licensed Clinical Social Worker (LCSW) actively engaged at Scripps Mercy Hospital Chula Vista, and drawing on my background in other health systems, direct clinical services, and social policy, I am acutely attuned to the formidable obstacles caregivers face when coordinating services and financing essential home care in the United States. The pressing need for innovative solutions like your proposed Olera CareNavigator cannot be overstated. The individuals under my care frequently lack the necessary resources and information to access vital home aid, transportation services, and financial assistance programs. This unfortunate reality often triggers a cycle of hospital readmissions, stemming from unmet needs due to prolonged service initiation. Your solution could significantly combat this problem and potentially disrupt this vicious cycle.

The potential of AI advancements is both captivating and promising, particularly in terms of increased ability to process resource data and in delivering empowering information to caregivers to lead to health behaviors. Given my experience, I am enthusiastic about endorsing and supporting the SBIR project, offering my expertise and shared passion for making a difference in the lives of those with dementia and disabilities.

The approach of reinforcing technology through human learning (RFHL) resonates logically to me and I agree is paramount to maintaining technological quality control. I am confident that my insights will be valuable in my contributions to the development of the AI serving on the RFHL advisory panel, aligning the technology with clinical social work standards, especially in its capacity to assess needs, identify resources, and facilitate empowering information dissemination. This is especially pertinent for family caregivers navigating progressive conditions like Alzheimer's dementia.

Furthermore, my advisory role extends to leveraging my expertise on local, state, and federal aid programs. My hands-on involvement in connecting families with underutilized aid initiatives, such as SSI, in-home caregiving, and extensions of Medicare/Medicaid, complements your initiative as I can help identify commonly underutilized programs for the team to target and add to your repository of resources. This Educational Materials and Caregiver Resources (EMCR) database could potentially be a game-changer if scaled, systematically mapping the "eldercare ecosystem" and standardizing nationwide data on care services, aid programs, and healthcare options. This database addresses a critical gap by providing reliable access to comprehensive information on resources for social workers like me.

Overall, this project resonates with my professional values and mission as a LCSW. If milestones laid out are successful, it could significantly reduce hospital overutilization by fostering a better-connected community with accurate information on resources and services and knowledge on how to initiate care, allowing more proactive resource utilization and better care over all for the elderly and those with disabilities like Alzheimer's dementia. I am excited to contribute my experience to pioneering advancements in this crucial field of research and supporting the aging population.

Sincerely,

Stephanie Jones, LCSW
Scripps Mercy Hospital Chula Vista

WALK TO END ALZHEIMER'S
ALZHEIMER'S ASSOCIATION

alz.org/walk

TJ Falohun
Co -Founder
Olera

We are thrilled to welcome you to the Walk to End Alzheimer's! On behalf of the entire organizing committee, we want to express our gratitude for your sponsorship commitment to join us in the fight against Alzheimer's disease. Your participation and dedication are invaluable in raising awareness and funds to support research, care, and support for those affected by this devastating illness.

This year, as we come together for the Walk, we are reminded of the strength and resilience of our community. Alzheimer's disease affects millions of individuals and their families, and it is through events like this that we can make a real difference. By walking side by side, we are not only raising funds but also showing our solidarity and support for those living with Alzheimer's and their caregivers.

The Walk to End Alzheimer's is more than just a fundraising event; it is a celebration of hope, unity, and determination. It is a day where we honor the memories of those we have lost, support those currently battling the disease, and inspire each other to keep pushing forward until we find a cure. Together, we can make a significant impact on the lives of those affected by Alzheimer's.

Your enthusiasm and dedication are crucial to achieving our goals as a team. We encourage you to engage your friends, family, and colleagues, and raise funds to support the Alzheimer's Association's vital programs and services. Every dollar raised brings us one step closer to a world without Alzheimer's.

Throughout the Walk, you can connect with others who share your passion and commitment. It is a chance to build new friendships, share stories, and find comfort in knowing that you are not alone in this fight. Together, we can create a supportive and compassionate community that uplifts and empowers each other.

We want to thank you once again for being part of this incredible journey. Your sponsorship will impact the lives of countless individuals affected by Alzheimer's disease. We cannot wait to see you at the Walk and witness the power of our collective efforts.

With heartfelt gratitude,

PV Villasenor-Sandell
Development Manger
210.963.5638 | pavillasenor@alz.org

SCHOOL OF PUBLIC HEALTH
HEALTH SCIENCE CENTER

Department Of Epidemiology & Biostatistics

TEXAS A&M UNIVERSITY

September 1, 2023

Olera Leadership,

I am excited to work with you on your SBIR application for the Olera CareNavigator project and share the same enthusiasm for its potential to address the critical needs of family caregivers, particularly in dementia caregiving. With extensive experience as a tenured Professor of Biostatistics at Texas A&M School of Public Health, I can bring in expertise in statistical design and analysis to the team. My background includes not only statistical theory but also its practical applications including the type of Alzheimer research proposed in this project, making me well-equipped to serve as the lead biostatistician to ensure the data collected in testing the technology is analyzed and reported appropriately and accurately.

I am also pleased to highlight my collaboration with Dr. Qiping Fan and Dr. Marcia Ory. Our collective expertise ensures a comprehensive and rigorous approach to the data analysis strategy, further strengthening the project's scientific foundation in technology usability research.

I am deeply committed to advancing the Olera CareNavigator project in collaboration with your team, and our colleagues at Clemson and Texas A&M. Together, we will work diligently to positively impact caregivers and those affected by dementia, report our findings, and eventually fulfil the project's promise to make a meaningful difference in lives of caregivers and care recipients. Please do not hesitate to contact me if you have any questions.

Sincerely,

Gang Han, Ph.D.
Professor of Biostatistics
School of Public Health
Texas A&M University
Tel: 979-436-9508
Email: gang.han@tamu.edu

210 SPH Administration Building (212 Adriance Lab Road)
1266 TAMU
College Station, TX 77843-1266

Tel. 979.436.9508; Fax 979.458.1877
ghan@sph.tamhsc.edu